DEAP EEG Emotion Recognition: A Comparative Analysis of Classical and Deep Learning Pipelines

Project Report: Methods, Results, Literature Review, and Findings

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Chapter 1

Introduction and Objectives

1.1 Purpose of this report

This document is a technical record of the work I carried out in this repository: a head-to-head comparison of two feature/modeling paradigms for binary emotion classification on the DEAP dataset, followed by an extended investigation into why subject-independent performance on this dataset is weak, and a final cross-corpus validation against SEED-IV. I wrote it so it can be expanded into a formal report or thesis chapter: the methods, the results, every literature source I consulted, and every open item are documented here in enough detail to reconstruct the reasoning behind each design decision.

1.2 Objectives

1. I built two parallel, directly-comparable pipelines for classifying DEAP's four emotion dimensions (valence, arousal, dominance, liking) from EEG: a classical-ML pipeline on hand-crafted Differential Entropy (DE) band-power features (Track A), and a deep-learning pipeline on raw, baseline-normalized EEG windows (Track B).

2. I evaluated both under a subject-independent protocol (`GroupKFold` / `GroupShuffleSplit`, grouped by subject) rather than the leakier, ungrouped splits common in the published literature.

3. Since subject-independent performance turned out to be weak, I tested plausible causes one at a time: feature richness, hyperparameter tuning, CV rigor, data quantity (window overlap), label definition, per-subject calibration, feature/distribution alignment, adversarial subject-invariance, self-supervised pretraining, an AUC-targeted loss, test-time adaptation, and diagnostic interpretability (SHAP).

4. I checked whether the resulting near-chance ceiling is a property of DEAP specifically, or of subject-independent EEG emotion decoding in general, via cross-corpus transfer to a second dataset (SEED-IV).

5. I grounded the design choices above in a review of 32 papers (`papers_EEG/`) and report every case where an intervention did not help.

1.3 How this report is organised

Chapter 2 describes the DEAP dataset and the two evaluation datasets involved (DEAP itself and SEED-IV). Chapter 3 maps the repository layout. Chapter 4 is the core of the report: the methods behind each experiment, the results, and a discussion, covering the main notebook

(`DEAP_Comparative_Analysis.ipynb`) end to end. Chapter 5 is an appendix documenting every standalone `scripts/*.py` file and the lower-level implementation detail (function names, cache file names) that the main chapter omits. Chapter 6 is the literature review and bibliography. Chapter 7 consolidates the numeric results into one master table. Chapters 9–12 cover what was accomplished, what was left undone, what I would do differently, and the concrete challenges I ran into. Chapter 13 concludes.

Chapter 2

Dataset and Problem Setup

2.1 DEAP

The DEAP dataset (Koelstra et al., 2012) (`data_preprocessed_python`) contains 32 subjects × 40 trials × 32 EEG channels sampled at 128 Hz. Each subject file `sXX.dat` is a pickled dictionary with:

- `data`: shape $(40, 40, 8064)$: 40 trials, the first 32 of 40 channels are EEG (the remaining 8 are peripheral physiological signals, dropped throughout this project). 8064 samples = 3 s pre-trial baseline (384 samples) + 60 s of stimulus (7680 samples).
- `labels`: shape $(40, 4)$: continuous 1–9 self-assessment ratings for valence, arousal, dominance, and liking.

Labels are binarised at the DEAP-standard threshold of 5.0 (rating $> 5 \Rightarrow$ class 1). Section 4.2 shows this binarisation is not balanced: valence is 21.0%/79.0%, arousal 23.3%/76.7%, dominance 60.9%/39.1%, and liking 66.5%/33.5% high/low, a fact that explains a large share of the most important finding in this report (Section 4.6).

2.2 SEED-IV

SEED-IV (Zheng et al., 2018) ("EmotionMeter") is used as the cross-corpus validation target in Section 4.10. It shares SEED's 62-channel montage and 15 subjects, but differs from the original 3-class SEED dataset in several respects that required a dedicated adapter (Section 4.10.2): 4 emotion classes (0 = neutral, 1 = sad, 2 = fear, 3 = happy) instead of 3; 24 film-clip trials per session instead of 15; 3 sessions per subject, each with its *own* trial-label order (unlike original SEED's single shared label file); and raw trials of variable duration (tens of thousands of samples at 200 Hz) rather than DEAP's fixed 60 s. I chose SEED-IV over original SEED because it is mirrored on Kaggle (`https://www.kaggle.com/datasets/phhasian0710/seed-iv`) without a signed data-use agreement, whereas original SEED, DREAMER, and AMIGOS all remain gated behind such agreements (see Chapter 10).

Chapter 3

Repository Structure

```
.
|-- README.md                          (dataset links, repo layout, quick start)
|-- Project_Report.tex / .pdf          (this report)
|-- DEAP_Comparative_Analysis.ipynb    (the main notebook)
|-- script_notebook_execution_map.txt (script<->notebook run-order/dependency map)
|-- citations.txt                      (flat paper citation list)
|-- literature_log.txt                 (per-paper attribution/summary log)
|-- deap-dataset\data_preprocessed_python\   (raw DEAP .dat files, 32 subjects)
|-- seed-dataset\                      (SEED-IV download: eeg_raw_data/{1,2,3}/, etc.)
|-- processed_cache\                   (cached .npz feature/window builds)
|-- results\
|   |-- tables\   (21 CSV result tables)
|   \-- figures\  (24 PNG figures)
|-- papers_EEG\   (32 literature-review PDFs, renamed to descriptive titles)
\-- scripts\
    |-- dl_models.py, rich_features.py
    |-- build_extended_features.py, build_overlap_windows.py, build_raw_ratings.py
    |-- track_a_*.py   (9 classical-ML companion experiments)
    |-- track_b_*.py   (5 deep-learning companion experiments)
    \-- cross_corpus\
        |-- base.py, seed_adapter.py, run_cross_corpus.py, README.md
```

The notebook is the primary deliverable; the `scripts/` directory holds every experiment that was too expensive to run inline, each of which writes its results to a CSV in `results/tables/`, which the notebook then reads and plots. This split is discussed further in Chapter 12. A handful of one-off scripts used to author the notebook's own cells during earlier stages of the project are not listed above; they are not part of the reproducible-results path. `script_notebook_execution_map.txt`, at the repository root, is the authoritative statement of that split: for every companion script it names the notebook section and cell range that consumes its output, the cache file it depends on, and the run order needed to reproduce every table and figure in this report from a clean checkout; Section 5.7 in the Appendix summarises it.

Chapter 4

Methods, Results, and Discussion

This chapter covers the main notebook (`DEAP_Comparative_Analysis.ipynb`) end to end: what I did, what I found, and what I make of it. It is organised by experiment, in the order I ran them, each with its own methods and results; a single Discussion section at the end synthesises all of them. Lower-level implementation detail, function names, and cache file names are in Appendix 5 rather than here.

Reproducibility: `SEED = 42` fixes randomness in scikit-learn/XGBoost/PyTorch/NumPy throughout. The notebook runs end to end without errors in about 16 minutes on the hardware described below.

4.1 Data and preprocessing

Data loading, feature extraction, and the deep-learning pipeline use `numpy`, `pandas`, `scipy.signal`, `scikit-learn` (Pedregosa et al., 2011) (`GroupKFold`/`GroupShuffleSplit`), `xgboost` (Chen & Guestrin, 2016), and `PyTorch` (Paszke et al., 2019), run on an NVIDIA GeForce RTX 3070 Ti. Windows are 4 s (`WINDOW_SEC = 4`), giving 15 non-overlapping windows per 60 s trial; the four spectral bands are theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz), and gamma (30–45 Hz); the label threshold is 5.0 on DEAP's 1–9 scale. All 32 subjects were used in every run. Both the window length and the band edges follow the conventions used across most of the DEAP emotion-recognition literature I reviewed (Chapter 6), which keeps the numbers in this report comparable to published work rather than an artefact of an idiosyncratic preprocessing choice: a wildly different window length or band definition would make any AUC comparison, including the leaky-split contrast in Section 4.11, an apples-to-oranges one.

Each subject's trial is first baseline-corrected: the per-channel mean of the 3 s pre-trial baseline is subtracted from the 60 s stimulus signal, removing subject/session DC offset and slow drift before either feature branch is computed. Skipping this step would let a classifier partly exploit each subject's idiosyncratic DC offset or electrode-impedance level rather than a genuine stimulus-driven signal, which would be a confound baked into every downstream number rather than a real finding. From the baseline-corrected signal, two parallel representations are built:

- **Differential Entropy (DE) features**, for Track A: a 4th-order Butterworth (Butterworth, 1930) zero-phase filter splits the signal into the four bands above, each band is cut into 15 non-overlapping 512-sample windows, and the Differential Entropy of each (channel, band) pair is computed as $\mathrm{DE} = 0.5 \log(2\pi e\, \sigma^2)$, the closed-form differential entropy of a Gaussian with the window's own variance σ^2. This gives 32 channels $\times$ 4 bands $=$ 128 features per window. I used Differential Entropy rather than raw band power because it is the dominant hand-crafted feature in the EEG emotion-recognition literature (Duan et al., 2013), shown in that original work to outperform simple band energy on exactly

this kind of task, which makes Track A a direct test of the field's default feature choice rather than a feature I invented for this project. A zero-phase (forward-backward) filter is used specifically so that filtering does not shift each band's energy in time relative to the other bands, which would otherwise distort the Differential Entropy computed from a fixed 512-sample window.

- **Raw windows**, for Track B: the same 512-sample windows kept in their raw $(32, 512)$ form and standardised per channel (zero mean, unit variance within that window) for stable CNN input. Per-channel standardisation matters here because raw EEG amplitude varies by an order of magnitude or more across subjects and recording sessions due to electrode impedance and skull thickness; without it, a convolutional filter's learned weights would need to simultaneously represent amplitude scale and signal shape, which empirically destabilises training far more than it does for vision-style inputs with a fixed, known dynamic range.

Looping this over all 32 subjects $\times$ 40 trials gives 19,200 windows total: a DE feature matrix, a raw-window tensor, one binary label array per task, and a subject-ID array used to group folds by subject so no subject's windows appear in both train and test. Results are cached to disk so repeated runs do not have to reprocess the raw files. Building both representations from the same baseline-corrected, windowed signal, rather than deriving Track B's windows independently, is what makes the Track A versus Track B comparison in Section 4.5 a comparison of modelling paradigm rather than of two different, incommensurable preprocessing pipelines.

4.2 Class balance

Class counts per task, computed directly from the window array: valence 21.0% high, arousal 23.3% high, dominance 60.9% high, liking 66.5% high. Valence and arousal are markedly imbalanced; dominance and liking are closer to balanced. This imbalance has to be checked before trusting any raw accuracy number, and it turns out to matter for most of the results in this chapter.

I ran this check, and put it before any model results rather than after, because accuracy alone is a misleading metric under class imbalance: a classifier that always predicts the majority class on valence would already score 79.0% accuracy while having learned nothing about the actual signal, and a reader who saw only an accuracy number for valence with no class-count context beside it would have no way to tell that classifier apart from a genuinely discriminative one. Khan et al. (2025) document exactly this failure mode recurring across published DEAP work, where headline accuracy numbers survive peer review without a class-balance caveat attached even though the underlying labels are known to be skewed; checking and reporting this up front is a direct response to that documented gap, not a routine formality. ROC-AUC, threshold-independent and insensitive to class proportions by construction, is used as the primary metric throughout the rest of this report for exactly this reason, with accuracy reported alongside it only as a secondary, easily-misread number. The practical consequence appears repeatedly later in this chapter: Track B's 83.5% valence accuracy (Section 4.4) sits within a few points of the 79.0% majority-class baseline established here, which is the first concrete instance of the accuracy/AUC divergence that Section 4.6 discusses at length and that recurs across nearly every high-capacity model in Chapter 8.

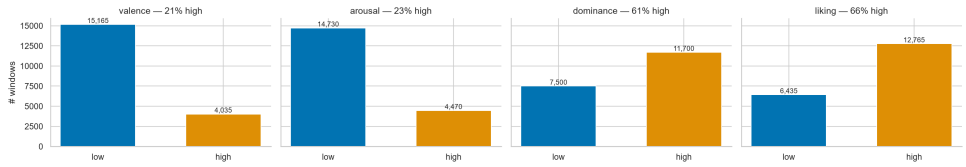


Figure 4.1: High/low class balance per task. Valence and arousal are markedly skewed (21.0%/79.0% and 23.3%/76.7% high); dominance and liking are closer to balanced.

4.3 Track A: classical models on DE features

Six models (Logistic Regression, kNN (Cover & Hart, 1967), Random Forest (Breiman, 2001), SVM-RBF (Cortes & Vapnik, 1995), XGBoost (Chen & Guestrin, 2016), MLP), each wrapped in a standard-scaling pipeline, evaluated with 5-fold `GroupKFold` (grouped by subject, so no subject's windows appear in both train and test). SVM-RBF's training folds are capped at 8000 samples, a pragmatic speed trade-off that does not apply to the other five models.

I chose these six specifically to span as many different inductive biases as a classical-ML comparison reasonably can on the same 128-dimensional Differential Entropy input: a linear hyperplane (Logistic Regression), a non-parametric distance vote with no training-time optimisation (kNN), two tree ensembles built from opposite objectives – bagging independent trees to reduce variance (Random Forest) versus boosting trees sequentially against residual error (XGBoost) – a kernel method that can represent a nonlinear decision boundary without an explicit feature map (SVM-RBF), and a generic trainable nonlinear function approximator (the MLP). The reason this breadth matters methodologically, not just descriptively, is made explicit in Chapter 8: because every one of these six model families, plus four structurally distinct deep architectures in Track B, converges on the same narrow AUC band, the breadth of model choice here is what rules out "a weak model family was picked" as an explanation for that finding. `GroupKFold`, grouped by subject rather than left ungrouped, is the single design choice most responsible for how low these numbers are relative to prior published work on this dataset (Section 4.11 quantifies the gap directly); it was chosen over an ungrouped split specifically because an ungrouped split lets windows from the same subject appear in both the training and test folds, which lets a classifier partly learn to recognise the subject rather than the emotion, inflating reported accuracy in a way that would not generalise to a genuinely new user. Table 4.1 shows the best model per task by ROC-AUC.

Table 4.1: Track A: best classical model per task (5-fold GroupKFold mean)

Task	Best model (by AUC)	Accuracy	F1 (macro)	ROC-AUC
valence	RandomForest	0.779	0.446	0.501
arousal	kNN	0.744	0.451	0.495
dominance	SVM-RBF	0.557	0.498	0.513
liking	SVM-RBF	0.587	0.517	0.534

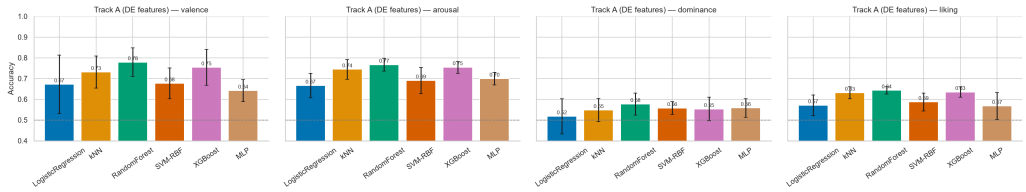


Figure 4.2: Track A: accuracy/F1/ROC-AUC for all six classical models across all four tasks (5-fold GroupKFold).

4.4 Track B: deep models on raw EEG windows

4.4.1 Model architectures

Four architectures, all implemented from scratch in PyTorch:

- EEGNet (Lawhern et al., 2018) (2130 params): temporal conv, then depthwise spatial conv, then separable conv, purpose-built for EEG.
- ShallowConvNet (Schirrmeister et al., 2017) (54,562 params): temporal conv, then spatial conv, then a square/log-power nonlinearity, inspired by FBCSP-style band-power features.
- CNN-LSTM (82,466 params): a spatial-temporal CNN front end collapses the channel dimension; a bidirectional LSTM then models the resulting temporal sequence.
- TSception (Ding et al., 2022) (8765 params): multi-scale temporal "inception" branches at three kernel sizes, fused with a global spatial conv and a hemisphere-asymmetry spatial conv using DEAP's standard channel order. This is a simplified reimplementation in the spirit of the original, not a line-for-line port of its codebase.

These four were chosen to mirror Track A's spread of inductive biases inside the deep-learning paradigm: EEGNet and ShallowConvNet are the two most-cited purpose-built EEG architectures in the literature I reviewed (Chapter 6), giving Track B a literature-grounded floor analogous to Track A's use of a standard hand-crafted feature; CNN-LSTM adds an explicit sequence-modelling component that neither purpose-built architecture has, to test whether within-window temporal dynamics carry information a purely spatial/spectral model would discard; and TSception is included specifically because it hard-codes a named neurophysiological hypothesis (hemisphere asymmetry) into its architecture, which lets Chapter 8 test whether encoding that hypothesis structurally helps, rather than assuming it does. Implementing all four from scratch, instead of importing each paper's reference implementation, was necessary because none of the original codebases targets this project's exact input shape (32 channels, 512 samples, DEAP's own channel ordering) or label set, and it makes every layer of every model auditable against the architecture described in its source paper rather than trusted as an opaque dependency.

4.4.2 Training procedure

Each model is trained under a subject-independent 70/15/15 train/val/test split by subject, so no subject's data crosses a split boundary. Training uses AdamW (Loshchilov & Hutter, 2019) with a plateau learning-rate schedule, automatic mixed precision on the GPU, and early stopping on validation loss (patience 8, max 40 epochs), restoring the best validation-loss checkpoint before final test evaluation. AdamW was chosen over plain Adam because its decoupled weight decay does not interact with the adaptive per-parameter learning rate the way Adam's coupled L_2 penalty does, which matters more than usual here given how small some of these networks are (EEGNet has only 2130 parameters): an entangled weight decay term can dominate the

update for a parameter with a large adaptive learning rate, effectively over-regularising exactly the filters a small network can least afford to lose capacity in. Early stopping on validation loss, rather than training for a fixed number of epochs, guards against overfitting to a training fold's particular mix of subjects; given how few distinct subjects populate each fold (Section 8.1), a model trained to convergence without this safeguard would have every opportunity to memorise subject-specific noise in the training set rather than a signal that generalises to the held-out subjects.

4.4.3 Results: all four models on all four tasks

16 training runs (4 architectures × 4 tasks), one train/val/test split each, on the GPU. Table 4.2 shows the best model per task.

Table 4.2: Track B: best deep model per task (single subject-independent split)

Task	Best model (by AUC)	Accuracy	F1 (macro)	ROC-AUC
valence	EEGNet	0.835	0.455	0.561
arousal	EEGNet	0.699	0.572	0.607
dominance	TSception	0.589	0.414	0.475
liking	CNN-LSTM	0.599	0.384	0.497

Dominance/EEGNet posts AUC 0.288, below chance and the single worst result in either track. It is an early instance of the arousal/dominance instability pattern that recurs throughout Sections 4.14 and 4.15.

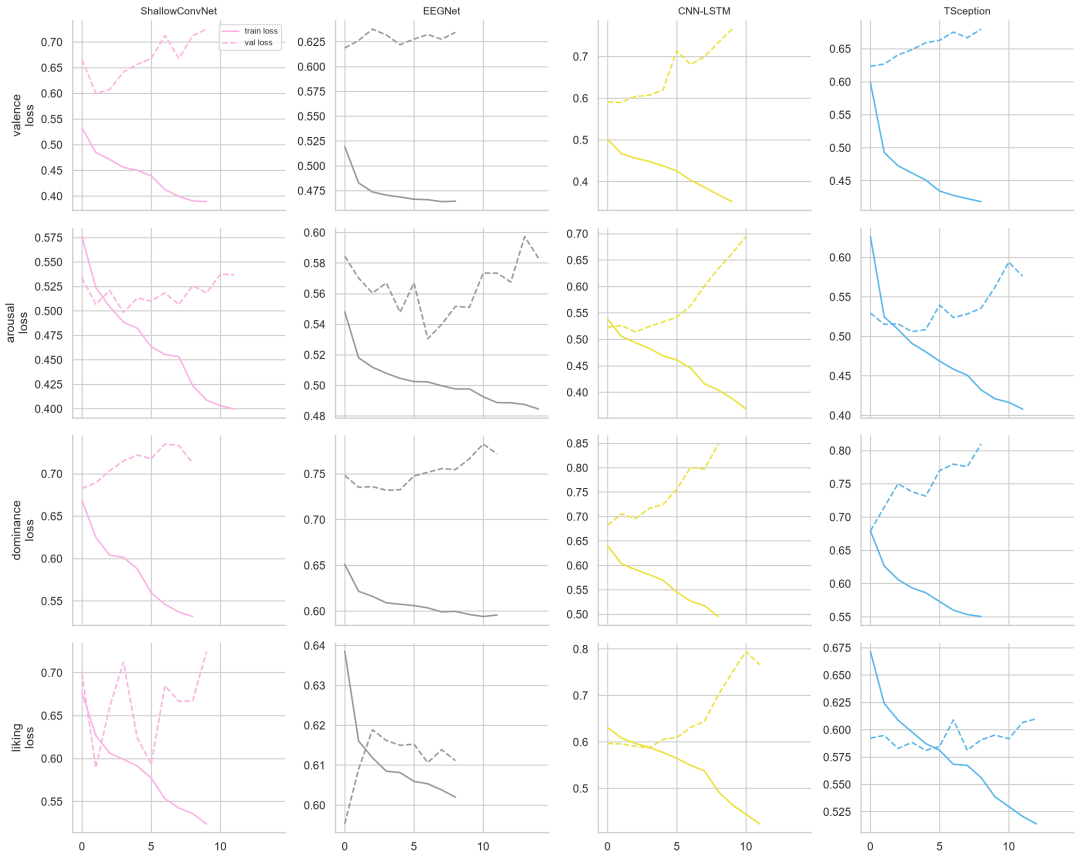


Figure 4.3: Track B: training/validation loss curves for all 16 runs (4 architectures × 4 tasks), showing early-stopping behaviour under the patience-8 schedule.

4.5 Combined comparative analysis

Track A and Track B results are concatenated into a single 10-model table and plotted side by side (accuracy/F1/AUC across all 10 models × 4 tasks), alongside confusion matrices for the model with the highest mean accuracy on each task.

This step exists because Sections 4.3 and 4.4 report Track A and Track B separately, which makes it easy to compare models within a track but not across the two paradigms the whole project is built to compare. Putting all ten models side by side on the same axes, for the same four tasks, is what turns two separate result sections into the direct answer to this report's central question (Chapter 4's opening objective): does hand-crafted-feature-plus-classical-ML or end-to-end deep learning perform better on subject-independent DEAP emotion classification? A reader should use this section's figures to check, for each task, which of the ten bars is tallest by AUC (not by accuracy, given Section 4.2's imbalance caveat), and whether that winner comes from Track A or Track B; Section 4.6 draws out what that comparison actually shows once class imbalance is accounted for.

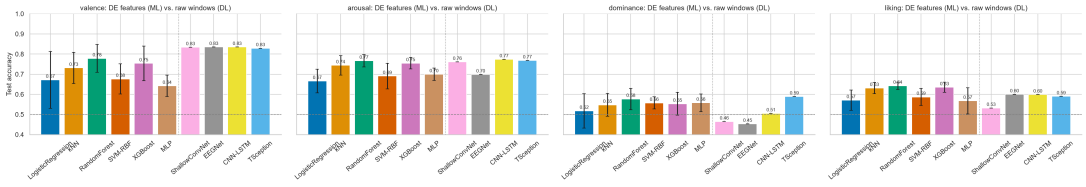


Figure 4.4: Combined accuracy/F1/ROC-AUC comparison across all 10 models (6 classical + 4 deep) and all 4 tasks.

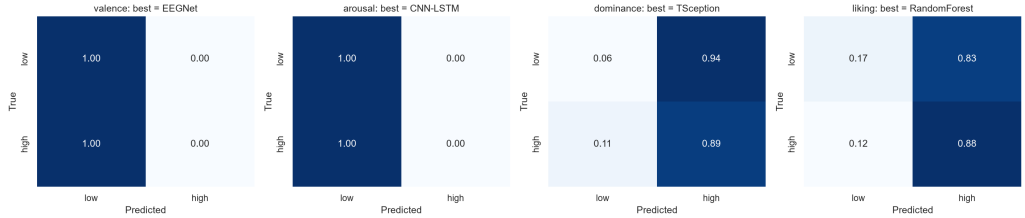


Figure 4.5: Confusion matrices for the highest-*accuracy* model on each task (EEGNet, CNN-LSTM, TSception, RandomForest). This selection is by mean accuracy across all 10 models, not by ROC-AUC, which is why the models named here differ from the AUC-based winners in Tables 4.1 and 4.2: it is a direct illustration of the accuracy/AUC divergence under class imbalance discussed in Section 4.6.

4.6 Discussion

A few points follow directly from the results above:

- Neither track wins outright. Track B edges out Track A on valence and arousal; Track A edges out Track B on dominance and liking. No model on either track clears about 0.61 AUC on any task: every one of the 10 models, on every one of the 4 tasks, falls in the 0.29–0.61 range (dominance/EEGNet the low outlier at 0.288, noted above; arousal/EEGNet the high end at 0.607), below chance at the bottom and barely above it at the top. This is the first appearance of a pattern the rest of this chapter spends many sections trying, and mostly failing, to explain away.

- The imbalance is not just a caveat, it shows up directly in the numbers. Valence is the most skewed task (79% majority class). Track B's valence models hit 83.5% accuracy at

only around 0.56 AUC: both numbers sit within a couple of points of the 79% majority baseline, the signature of a classifier leaning on class imbalance rather than a real signal.

- Compute cost differs a lot between tracks. Track A's cheapest models (kNN, XGBoost, Logistic Regression) are 10–1000× cheaper per fit than any Track B model, and Track A's numbers are 5-fold cross-validated means, while Track B's are single-split point estimates (Section 4.9 later adds repeats).
- Subject independence is the hard setting. Section 4.11 quantifies how much harder: my subject-independent AUC (0.288–0.607 across all ten models) sits far below the 0.97 ROC-AUC reported on the same dataset by Hasan & Islam (2025).

4.7 Richer Track A features

This section tests the most obvious candidate explanation for Section 4.6's near-chance ceiling: that a 128-dimensional Differential Entropy vector is simply too impoverished a feature set, and a richer, more diverse description of the same signal would let a classical model find the signal DE alone misses. I added three additional feature families to the 128-dim DE baseline (295 features total across 19,200 windows), each targeting a different kind of structure DE cannot represent by construction, since DE only summarises each (channel, band) window's variance and discards everything else about its shape, timing, and relationship to other channels:

- Hjorth parameters (activity, mobility, complexity) (Hjorth, 1970): 96 features (3×32 channels), time-domain complexity descriptors that DE alone misses, since they summarise a signal's variance, mean frequency, and frequency spread directly from its time-domain derivatives rather than from a band-limited variance.
- Frontal/hemispheric alpha asymmetry (FAA): 14 features, $\text{alpha}_{\text{right}} - \text{alpha}_{\text{left}}$ per electrode pair, the classic Davidson-style valence marker (Davidson, 1992), which predicts that positive-valence states are associated with relatively greater left-frontal alpha suppression (i.e. greater left-frontal activation) than negative-valence states. It is included specifically because it is the one feature family in this report built around a named, falsifiable neuroscience hypothesis rather than a generic signal descriptor, which lets Section 4.15.5's SHAP diagnostic and Chapter 8's TSception analysis test that specific hypothesis directly.
- Connectivity (PLV and coherence): 57 features across 19 curated channel pairs, phase-locking value (Lachaux et al., 1999) in the alpha and beta bands (38 features) plus alpha-band magnitude-squared coherence (19 features), computed via a Hilbert-transform/Welch-style pipeline. Both measures target inter-channel synchrony rather than per-channel power, on the premise that emotional state may be encoded more in how regions communicate than in how much power any single region produces.

4.7.1 Feature-set ablation

6 feature-set combinations × 3 representative models (Logistic Regression, Random Forest, XGBoost) × 4 tasks × 5-fold `GroupKFold` gives 360 fits. Mean AUC across all six feature sets ranges only from 0.478 (DE+Hjorth) to 0.494 (DE+Connectivity): every individual model/task fit stays in a 0.42–0.56 range, similar to the classical (non-deep-learning) models in Sections 4.3–4.4 but narrower than those sections' full 0.29–0.61 spread once Track B's deep models are included. None of the three additions move AUC meaningfully off chance for any task. I carry the "Full" set forward to Section 4.8 as the most complete configuration, not because the ablation crowned it a clear winner.

Table 4.3: Feature-set ablation: mean ROC-AUC across 3 models × 4 tasks (12 fits) per feature set

Feature set	#Features	Mean ROC-AUC
DE only	128	0.479
DE + Hjorth	224	0.478
DE + FAA	142	0.483
DE + Connectivity	185	0.494
DE + Hjorth + FAA	238	0.479
Full (DE+Hjorth+FAA+Conn)	295	0.483

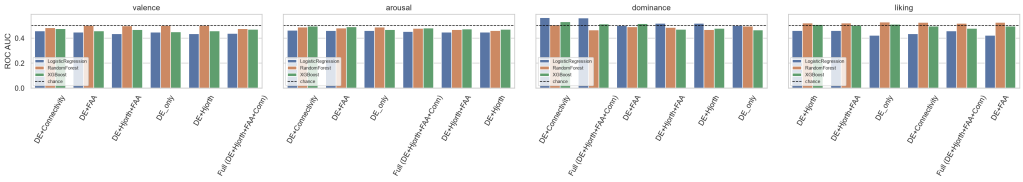


Figure 4.6: Feature-set ablation: ROC-AUC by feature set, model, and task.

4.8 Repeated and nested GroupKFold

Two upgrades here:

1. Repeated GroupKFold (5 repeats × 5 folds = 25 folds per model/task), giving every AUC a real 95% confidence interval. Since `scikit-learn` has no repeated-group variant, each repeat randomly relabels the 32 subject IDs before calling `GroupKFold(5)`, because its greedy fold assignment is sensitive to the sorted order of group labels, so relabeling yields a different subject-to-fold partition each repeat.

2. Nested CV for XGBoost and Random Forest: outer `GroupKFold(5)`, inner `GroupKFold(3)` via a randomized hyperparameter search (8 draws), to check whether the fixed, hand-picked hyperparameters used everywhere else were leaving accuracy on the table.

Of 24 model/task 95% AUC confidence intervals, 15 contain 0.5 (chance). The 9 that do not skew mostly toward dominance and liking (dominance/Logistic Regression: 0.571, CI [0.546, 0.597]; liking/kNN: 0.532, CI [0.518, 0.546]). Nested hyperparameter search does not close the gap either: dominance/XGBoost reaches 0.494 tuned versus 0.473 with the fixed hyperparameters used elsewhere in this report (Table 8.1), a small +0.021 change that is well within fold-to-fold noise. Table 4.4 gives the full picture across both tuned models and all four tasks, not just that one cell: five of the eight tuned/fixed deltas are negative, and the largest positive delta (dominance/XGBoost, +0.021) is barely larger than the largest negative one (arousal/RandomForest, −0.035), so there is no task on which tuning produces a change that looks different in kind from fold-to-fold noise. Together, these results rule out fixed hyperparameters as the main bottleneck.

Table 4.4: Nested GroupKFold hyperparameter search: tuned vs. fixed-hyperparameter ROC-AUC, RandomForest and XGBoost, all four tasks

Task	Model	Fixed HP AUC	Tuned AUC	Δ
valence	RandomForest	0.501	0.492	−0.009
valence	XGBoost	0.455	0.477	+0.022
arousal	RandomForest	0.489	0.454	−0.035
arousal	XGBoost	0.483	0.470	−0.013
dominance	RandomForest	0.497	0.489	−0.008
dominance	XGBoost	0.473	0.494	+0.021
liking	RandomForest	0.528	0.524	−0.004
liking	XGBoost	0.505	0.484	−0.021

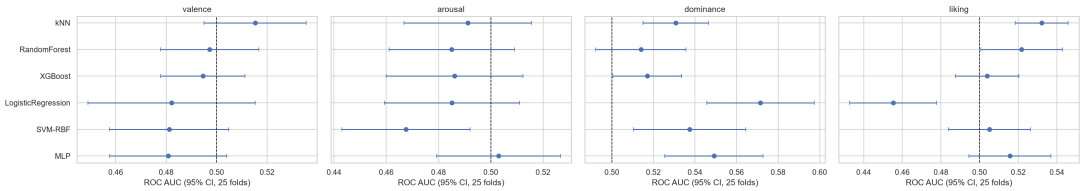


Figure 4.7: Repeated GroupKFold (5×5=25 folds): ROC-AUC with 95% confidence intervals per model/task; 15/24 intervals contain chance (0.5).

4.9 50% overlapping windows and repeated Track B splits

I built a second raw-window cache with 2 s stride (50% overlap) instead of the original 4 s stride, giving 29 windows/trial instead of 15 (37,120 windows total). This comes with a leakage caveat worth stating up front: overlapping windows share up to 50% of their raw samples and are not independent observations. Subject grouping still prevents cross-subject leakage, but not this within-subject correlation. I then ran 5 repeated 70/15/15 subject-independent splits, on both window variants, across all 4 architectures and all 4 tasks (160 training runs, about 87 minutes on the project's RTX 3070 Ti). Mean AUC across all model/task combinations is 0.483 (non-overlap) versus 0.498 (overlap50): the deltas are small and inconsistent in sign, consistent with the non-independence caveat. More correlated windows is not the same as more information.

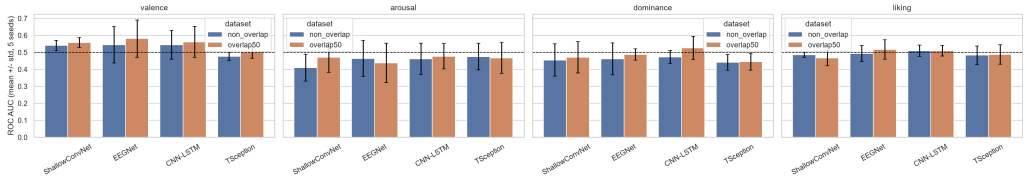


Figure 4.8: Repeated Track B splits: ROC-AUC, non-overlapping vs. 50%-overlap windows, across 4 architectures × 4 tasks.

4.10 Cross-corpus validation (SEED-IV)

4.10.1 Why SEED-IV, not original SEED

Original SEED requires a signed BCMI data-use agreement; DREAMER and AMIGOS require similar gated requests. SEED-IV, a related 4-class dataset from the same lab, is mirrored on

Kaggle with no agreement required, so it is the one target corpus I could actually obtain and run.

4.10.2 Pipeline

The cross-corpus code has three parts:

- Dataset-agnostic infrastructure: channel-name intersection between DEAP and a target montage, polyphase resampling, generic band-DE extraction restricted to the common channel subset, rebuilding DEAP's own DE features on that same restricted/reordered channel set so the DEAP-side model is retrained fairly rather than fed a mismatched feature vector, and zero-shot evaluation with a per-subject breakdown.
- A SEED-IV adapter that loads the 62-channel, 200 Hz session files, resolves the channel-order file defensively across several candidate filenames (the Kaggle mirror ships a differently-named and differently-capitalised file than originally assumed), and resolves per-session labels defensively: it first tries a real label file, and, since the download ships no separate label file for this purpose, only prose in a README, falls back to the label sequence published in the SEED-IV paper. I checked that fallback against the README by hand and it matches. SEED-IV's 4 classes are then mapped onto a valence-like binary task: happy is positive, sad/fear is negative, neutral is dropped.
- An orchestration script that runs (1) a within-DEAP `GroupKFold` baseline on the 32 common channels shared with SEED-IV, (2) DEAP→SEED-IV zero-shot transfer (train on all of DEAP, never see SEED-IV), and (3) a SEED-IV-only LOSO baseline (15 subjects, 15 folds, one per SEED-IV subject).

4.10.3 Result

Table 4.5: Cross-corpus validation: DE + XGBoost, 32 common channels

Setting	Accuracy	ROC-AUC	Notes
Within-DEAP GroupKFold	0.756	0.452	below chance point estimate
DEAP→SEED-IV zero-shot	0.709	0.517	marginally above chance
SEED-IV-only LOSO	0.709	0.673	14/15 subjects >0.55 AUC

Table 4.6 gives the full per-subject breakdown behind the SEED-IV-only LOSO row above: 14 of the 15 held-out subjects clear 0.55 AUC; the single exception is subject 97132107 at 0.515, still close to chance but not below it. There is no subject whose result is so far below chance that it would be worth excluding as an anomaly; the spread (0.515–0.880) is wide but one-sided, consistent with the pipeline finding a real, if uneven, signal on this corpus rather than noise centred on 0.5.

Table 4.6: SEED-IV-only Leave-One-Subject-Out: per-subject ROC-AUC (DE + XGBoost, 32 common channels)

Held-out subject	Accuracy	F1 (macro)	ROC-AUC
10230076	0.833	0.795	0.880
17741077	0.808	0.737	0.796
84163914	0.734	0.539	0.744
43091099	0.742	0.682	0.759
23084164	0.784	0.648	0.720
48942731	0.737	0.626	0.707
59416627	0.716	0.649	0.673
35646451	0.717	0.565	0.670
99723240	0.675	0.563	0.663
61771259	0.710	0.438	0.609
82638880	0.645	0.574	0.621
43051602	0.714	0.462	0.595
75903953	0.452	0.452	0.560
59481326	0.656	0.547	0.582
97132107	0.709	0.441	0.515

This changes how I read the question this section set out to answer. DEAP's own within-corpus signal is not inflated relative to cross-corpus transfer; it is the weakest of the three settings here. The identical pipeline reaches a moderate AUC of 0.673 on SEED-IV under a directly comparable protocol. That is one 3-setting comparison with a single model on a dataset with a different label construct (a 4-class emotion label collapsed to a happy-vs-sad/fear binary task, rather than DEAP's continuous valence threshold), so it supports the more modest claim that the near-chance ceiling may be specific to DEAP, rather than ruling out a general explanation for weak subject-independent EEG emotion decoding. Other caveats: the two corpora differ in session count and per-trial duration, and restricting to 32 common channels discards SEED-IV's denser 62-channel coverage.

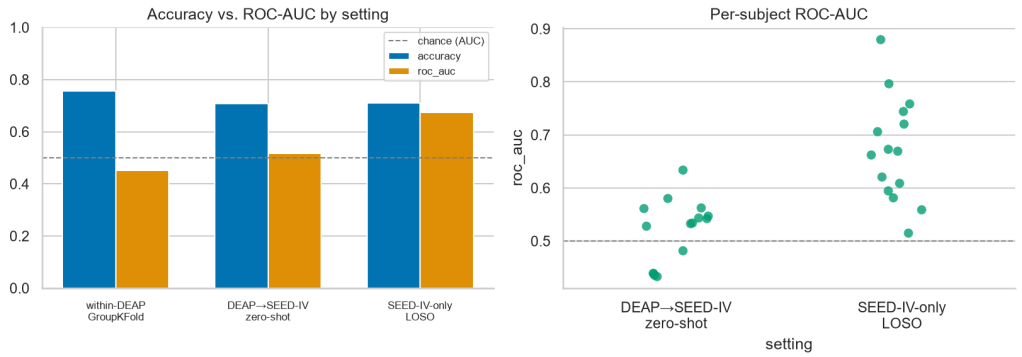


Figure 4.9: Cross-corpus validation: ROC-AUC across the three settings of Table 4.5 (within-DEAP, DEAP→SEED-IV zero-shot, SEED-IV-only LOSO).

4.11 Extended discussion

Pulling together Sections 4.7–4.10: AUC hovers within noise of chance almost everywhere, and this is the single most important finding in this report. None of richer features, repeated/nested CV, or window overlap fixed the ceiling, which is itself informative: each rules out a specific,

plausible cause rather than pointing to a failed implementation. The cross-corpus result overturns my own original guess that a weak DEAP signal would mean weak transfer too, and instead suggests the ceiling may be specific to DEAP rather than universal (Section 4.10). The practical recommendation I would make for any paper drawn from this work is to lead with the leaky-split contrast: my own 0.288–0.607 AUC under leakage-free `GroupKFold`, versus the 0.97 ROC-AUC reported on the same dataset and a similar feature/classifier combination by Hasan & Islam (2025).

4.12 Label-cleanup ablation

I tested three ways of turning continuous ratings into binary labels (Full feature set, 3 models, 4 tasks, `GroupKFold(5)`):

- raw threshold (baseline): fixed cutoff at 5.0.
- per-subject z-score: z-score each subject's own 40 ratings before thresholding at 0, correcting for individual anchoring bias.
- drop-middle: keep only ratings ≤ 3 or ≥ 6, discarding the ambiguous middle.

Mean AUC is 0.483 (raw threshold), 0.514 (per-subject z-score), 0.487 (drop-middle). Per-subject z-scoring rebalances classes substantially (valence goes from 21.0% to 47.3% high) and gives the largest AUC bump, but the gain is modest, and the class-balance shift means its accuracy numbers should not be compared directly to the baseline's. Neither alternative beats the baseline by enough to conclude that the fixed threshold was the main bottleneck.

Table 4.7: Label-cleanup ablation: mean ROC-AUC across 3 models × 4 tasks per label variant

Label variant	Mean ROC-AUC
raw_threshold (baseline)	0.483
per_subject_z	0.514
drop_middle	0.487

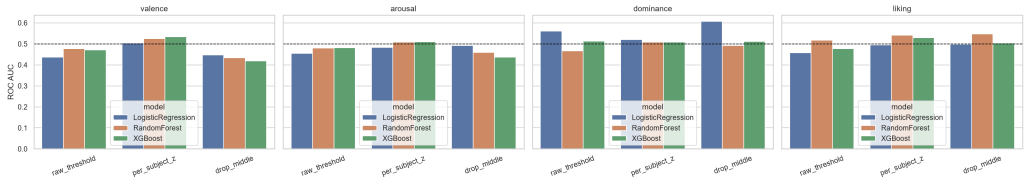


Figure 4.10: Label-cleanup ablation: ROC-AUC by label variant, model, and task.

4.13 Per-subject calibration (personalisation)

I ran a true 32-fold Leave-One-Subject-Out with XGBoost, which is cheap enough to make full LOSO tractable (about 13 s/subject). For each held-out subject: train a base model on the other 31 subjects; split the held-out subject's 40 trials 80/20 into eval/calibration sets, by trial, so a trial's 15 windows never cross the split; score the base model zero-shot on the eval set; then warm-start about 40 extra boosted trees on the 8-trial calibration set only, and re-score on the identical eval set. Subjects whose calibration set is single-class (13 across all 4 tasks) are skipped for the calibrated arm and reported as such. Mean zero-shot AUC is around 0.50–0.51 across all four tasks; mean calibrated AUC rises to 0.577 (arousal), 0.519 (dominance), 0.539

(liking), 0.537 (valence), a modest positive shift from a small amount of the target subject's own data. This suggests the ceiling is not purely a between-subject-variability problem that a little calibration data cannot touch, though 8 trials and 40 trees is a small intervention.

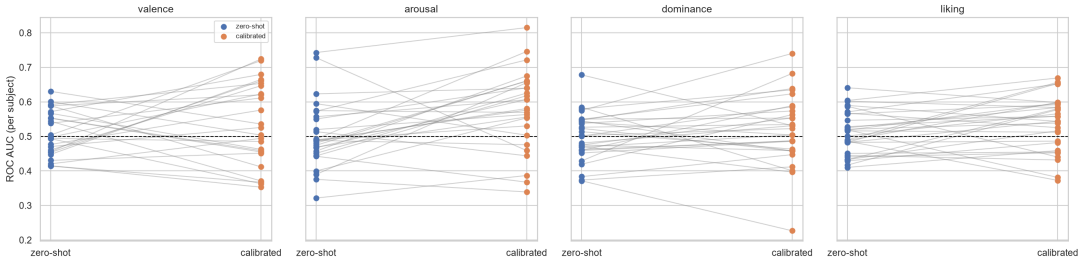


Figure 4.11: Per-subject LOSO calibration: zero-shot vs. calibrated ROC-AUC for each of the 32 held-out subjects, per task.

4.14 Literature-driven interventions

32 papers in `papers_EEG/` were assembled and reviewed (see Chapter 6), yielding three untried interventions.

4.14.1 Permutation significance test

I shuffle labels at the trial level within each subject (each subject keeps its own true set of 40 trial labels; only which of that subject's trials gets which label is permuted), then propagate each shuffled trial label to all 15 windows of that trial, rerun the exact Full-feature-set `GroupKFold(5)` pipeline with Logistic Regression as a fast proxy, and repeat this 300 times per task to build an empirical null AUC distribution; $p = \frac{1+\#\{\text{null} \geq \text{observed}\}}{n_{\text{perm}}+1}$. Permuting at the trial level rather than the window level matters: an earlier version of this test shuffled all 19,200 window-level labels independently, which treats a trial's 15 windows as exchangeable when they in fact share one label. That inflated the null's effective sample size and collapsed its spread (std ≈ 0.005), which in turn inflated every z-score, including dominance's. Once permutation respects the trial/subject structure, the null is roughly five times wider (std ≈ 0.022–0.028), and its mean is free to drift away from 0.5 per task (e.g. arousal's null mean is 0.532) since each permutation still preserves each subject's real, imbalanced trial counts. Under this corrected test, valence, arousal, and liking's observed AUC is no longer distinguishable from the null ($z = -2.25, -2.90, -1.12$; $p = 0.993, 0.997, 0.877$), so the earlier claim that they sit significantly *below* chance does not survive the correction. Dominance remains the only task whose observed AUC clears the null (observed 0.561 vs. null 0.499 ± 0.028, $z = +2.18$, $p = 0.0199$), but the effect has shrunk sharply from the buggy version ($p = 0.0033 \rightarrow p = 0.0199$) and does not survive a Bonferroni correction for testing four tasks ($\alpha = 0.05/4 = 0.0125$). The honest reading of the corrected test is that no task's AUC is robustly distinguishable from a properly-built, trial-respecting chance baseline.

Table 4.8: Permutation significance test (300 shuffles per task, trial-within-subject permutation, LogisticRegression proxy)

Task	Observed AUC	Null mean ± std	z	p
valence	0.437	0.495 ± 0.026	-2.25	0.9934
arousal	0.454	0.532 ± 0.027	-2.90	0.9967
dominance	0.561	0.499 ± 0.028	$+2.18$	0.0199
liking	0.458	0.482 ± 0.022	-1.12	0.8771

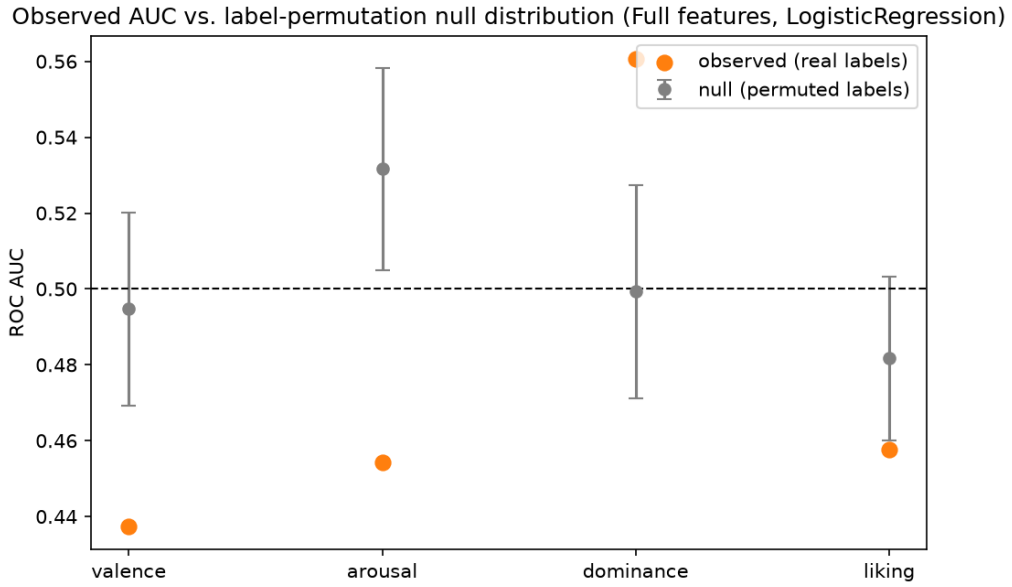


Figure 4.12: Permutation null AUC distributions vs. observed AUC, per task.

4.14.2 CORAL feature alignment

I implemented CORAL (Sun & Saenko, 2016): per `GroupKFold` fold, whiten the training fold's features by their own covariance and recolour by the unlabelled test fold's covariance, via eigendecomposition-based matrix square roots with a ridge for numerical stability. Mean AUC delta across 12 task/model combinations is +0.0059 (8/12 improved, 4/12 worse), small and inconsistent in sign.

Table 4.9: CORAL alignment: mean ROC-AUC across 3 models per task

Task	No alignment	CORAL	Δ
valence	0.462	0.478	+0.016
arousal	0.472	0.476	+0.004
dominance	0.514	0.521	+0.008
liking	0.484	0.480	−0.004

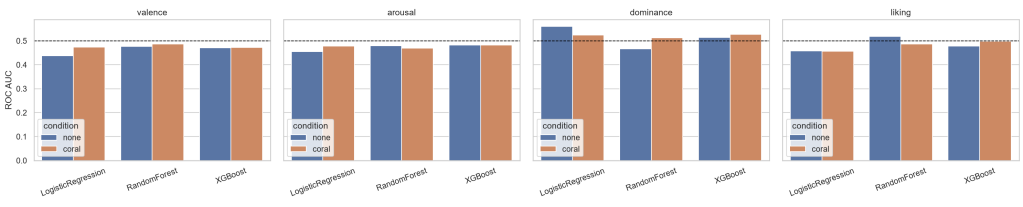


Figure 4.13: CORAL alignment: ROC-AUC before/after, by model and task.

4.14.3 Subject-adversarial (DANN) training

A note on the EEGNet baseline used from here on: Table 4.2 reports EEGNet's valence AUC as 0.561, from the main notebook's single 70/15/15 split. Sections 4.14.3 through 4.15 instead use 0.474 as the EEGNet valence "no intervention" baseline. Both numbers are correct for what they measure: they come from separate script runs with a different train/val/test split and seed than the main notebook cell, since each companion script draws its own split rather than reusing

the notebook's. I have not reconciled the two into a single number; they should be read as two separate point estimates of the same quantity rather than a single ground truth.

I added a gradient-reversal-layer (GRL) domain classifier to EEGNet, trained to predict the training-fold subject's identity; the GRL reverses that gradient before it reaches the shared encoder, using the standard Ganin & Lempitsky (2015) progressive-λ schedule. The outcome is mixed and task-dependent. Valence AUC 0.474 $\rightarrow$ 0.513 (+0.039) and liking AUC 0.467 $\rightarrow$ 0.517 (+0.050) improved; arousal AUC 0.616 $\rightarrow$ 0.400 ($-$0.216) and dominance AUC 0.552 $\rightarrow$ 0.403 ($-$0.149) collapsed, with precision and recall pinned near 0.50, the classifier collapsing toward one class rather than a milder generic drop. This matches a documented failure pattern in unsupervised domain adaptation, over-aggressive alignment collapsing class structure, described in one of the survey papers in the literature corpus (Sreehari et al., 2026).

Table 4.10: Subject-adversarial (DANN) training: ROC-AUC, EEGNet, single split

Task	Baseline	DANN	Δ
valence	0.474	0.513	+0.039
arousal	0.616	0.400	$-$0.216
dominance	0.552	0.403	$-$0.149
liking	0.467	0.517	+0.050

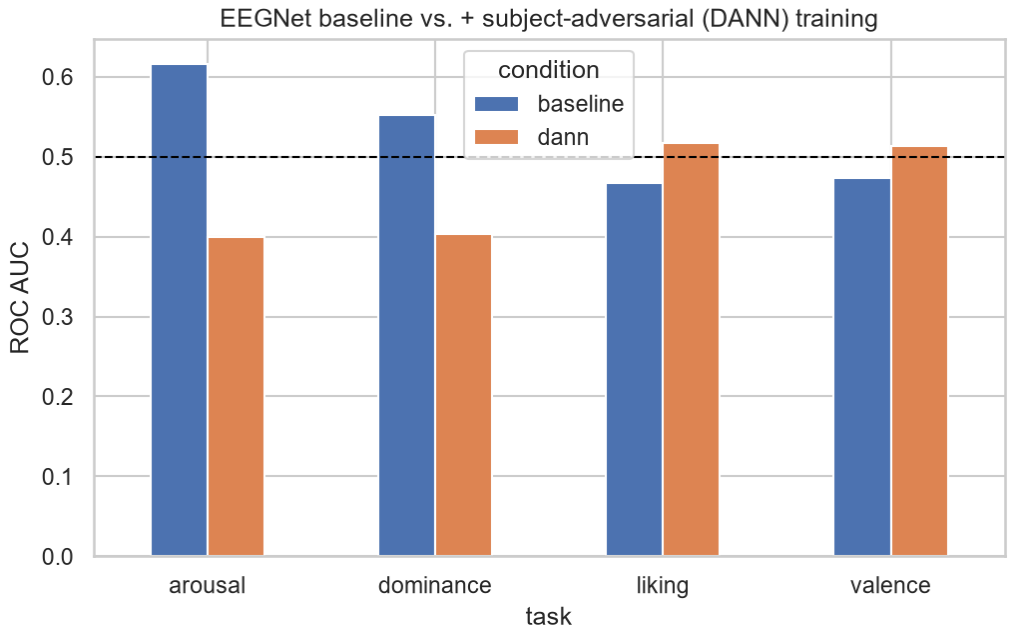


Figure 4.14: DANN training: ROC-AUC before/after, per task; note the arousal/dominance collapse.

4.15 Five deferred literature techniques

4.15.1 Euclidean Alignment

I implemented Euclidean Alignment, EA (He & Wu, 2019): per subject, compute each window's raw spatial covariance $C_i = X_i X_i^\top / T$, average across that subject's windows to get $\bar{R}$, then whiten every window by $\bar{R}^{-1/2}$ before DE features are recomputed. This is an upstream, per-subject, label-free, raw-channel-level transform, unlike CORAL's downstream per-fold feature-level operation. This is the strongest effect I found in the whole project: mean AUC delta +0.044

across 12 task/model combinations (10/12 improved), and unlike every other alignment technique I tried, the sign is consistent within three of the four tasks (valence +0.087/+0.027/+0.069, arousal +0.078/+0.027/+0.065, dominance +0.074/+0.038/+0.073 across LR/RF/XGBoost). It is still modest in absolute terms: the best post-EA AUCs land around 0.53–0.58.

Table 4.11: Euclidean Alignment: mean ROC-AUC across 3 models per task

Task	No alignment	EA	Δ
valence	0.467	0.528	+0.061
arousal	0.473	0.530	+0.057
dominance	0.488	0.550	+0.062
liking	0.488	0.485	−0.003

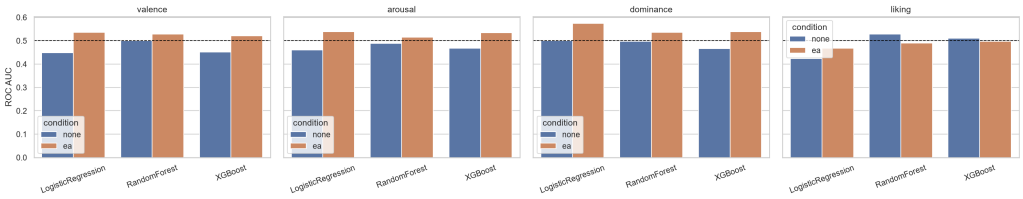


Figure 4.15: Euclidean Alignment: ROC-AUC before/after, by model and task, the strongest effect found in the project.

4.15.2 Self-supervised (TFR-style) pretraining

I pretrained EEGNet's conv trunk with a label-free MSE regression pretext task, predicting each window's own 32-channel × 4-band log-band-power from the raw signal using training-fold subjects only, then fine-tuned end to end and compared against an identically trained scratch EEGNet. The result is a bimodal, task-split pattern: valence AUC 0.474 → 0.547 (+0.074) and liking 0.467 → 0.567 (+0.100) improved, while arousal 0.616 → 0.534 (−0.082) and dominance 0.552 → 0.500 (−0.053) got worse. The pretext task itself converged fine, with MSE dropping over 15 epochs, so this is not a training-failure artifact.

Table 4.12: SSL pretraining: ROC-AUC, EEGNet, single split

Task	Scratch	SSL-pretrained	Δ
valence	0.474	0.547	+0.074
arousal	0.616	0.534	−0.082
dominance	0.552	0.500	−0.053
liking	0.467	0.567	+0.100

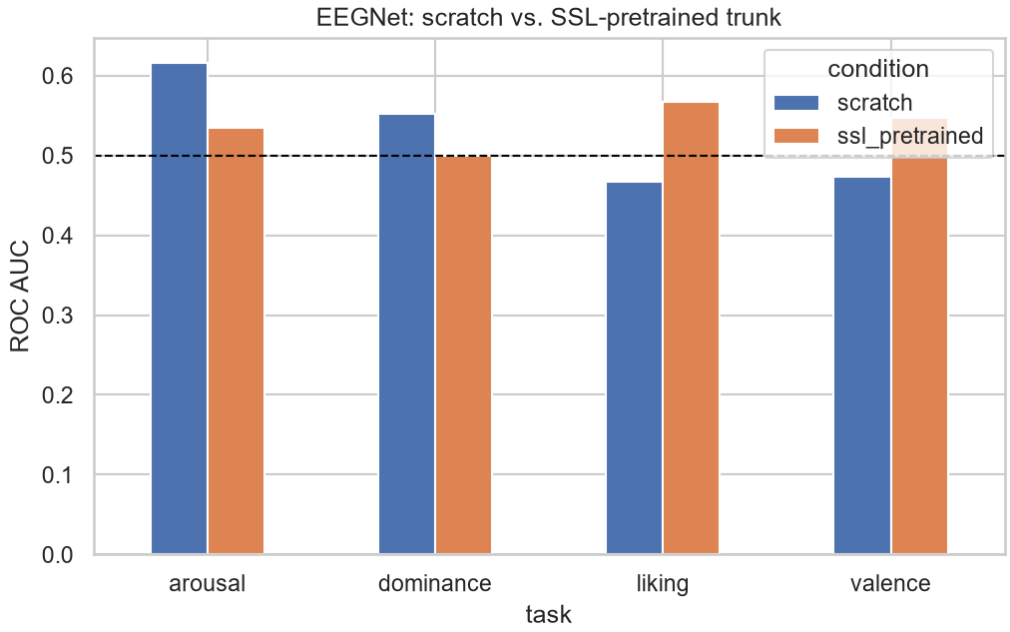


Figure 4.16: SSL pretraining: ROC-AUC, scratch vs. pretrained, per task.

4.15.3 AUC-margin training loss

I replaced cross-entropy with a mini-batch pairwise squared-hinge AUC surrogate, $L = \text{mean}_{i \in \text{pos}, j \in \text{neg}} \max(0, m - (s_i - s_j))^2$, a simplified, self-contained version of the Xiao (2024) AUC-maximization objective, not the source paper's full saddle-point solver. This is the one intervention with a clear mechanism behind it. Valence improved most (AUC 0.474 → 0.557, +0.083): the cross-entropy baseline's 83.5% accuracy at 0.474 AUC was a majority-class collapse, and the AUC-margin loss produced a more balanced classifier (accuracy 0.507, AUC 0.557) because it is insensitive to that imbalance by construction. Dominance improved slightly (+0.008); arousal (−0.010) and liking (−0.018) were small changes, likely within noise.

Table 4.13: AUC-margin loss: ROC-AUC, EEGNet, single split

Task	CE baseline	AUC-margin	Δ
valence	0.474	0.557	+0.083
arousal	0.616	0.606	−0.010
dominance	0.552	0.560	+0.008
liking	0.467	0.450	−0.018

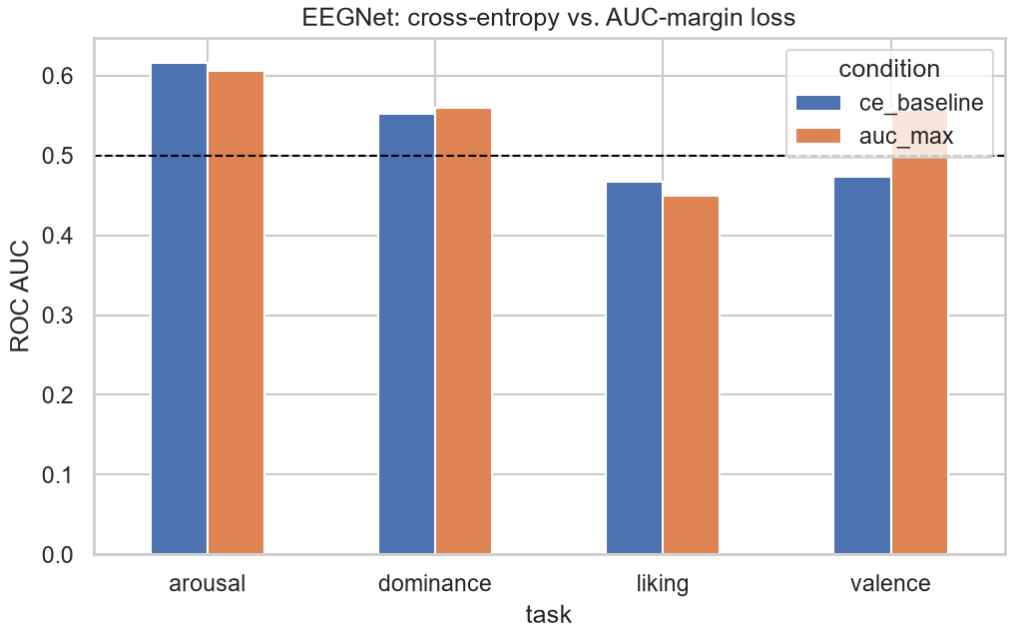


Figure 4.17: AUC-margin loss: ROC-AUC, cross-entropy vs. AUC-margin, per task.

4.15.4 AdaBN test-time adaptation

I recomputed every BatchNorm layer's running statistics using only a held-out test subject's own unlabelled windows, with no gradients and no labels, then evaluated with the adapted statistics. The net effect is negative overall. Arousal AUC 0.616 → 0.515 (−0.101) and dominance 0.552 → 0.489 (−0.063) got worse; liking 0.467 → 0.549 (+0.082) improved; valence was flat (+0.002). This is the only clearly net-negative intervention in Section 4.15.

Table 4.14: AdaBN test-time adaptation: ROC-AUC, EEGNet, single split

Task	No adaptation	AdaBN	Δ
valence	0.474	0.476	+0.002
arousal	0.616	0.515	−0.101
dominance	0.552	0.489	−0.063
liking	0.467	0.549	+0.082

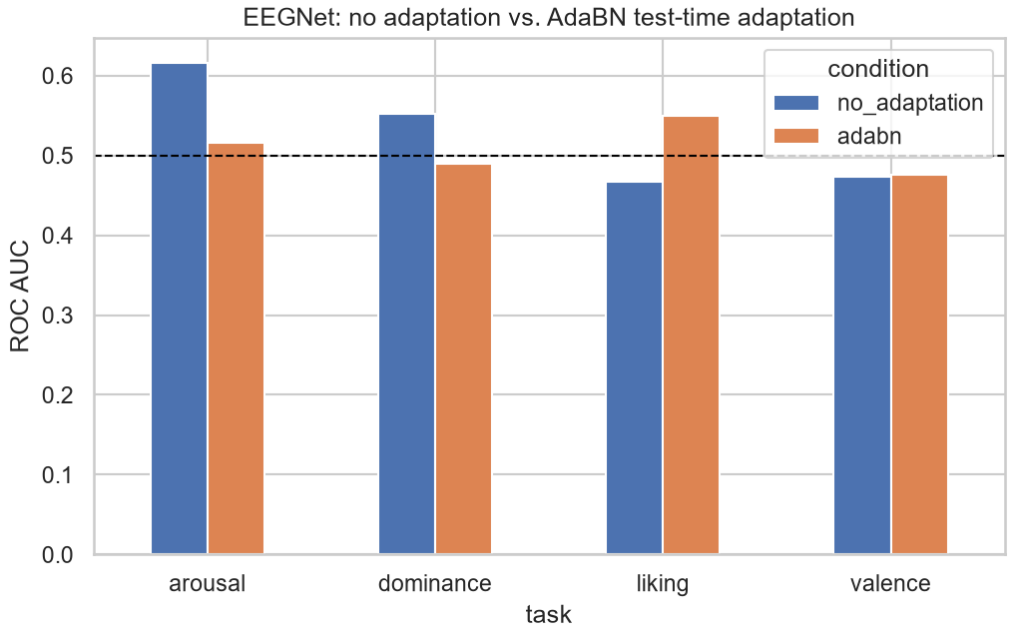


Figure 4.18: AdaBN test-time adaptation: ROC-AUC before/after, per task.

4.15.5 SHAP interpretability diagnostic

I fit XGBoost on the Full feature set, one representative fold per task, and computed SHAP values on the held-out fold, aggregated by feature family and individually. Importance is spread across families roughly in proportion to how many features each contributes, with two mild departures: DE's share (47–52%) and Connectivity's (22–24%) both sit slightly above their feature-count share, while Hjorth's (21–26%) sits below. Frontal/hemispheric alpha asymmetry, the one feature family built around a named neurophysiological hypothesis, carries the smallest importance share in every task (3.4–6.1%). The top individual features are diffuse spectral and connectivity structure (`de_beta_P4` for valence, `de_gamma_FC2` for arousal and liking, `plv_beta_F3-O1` for dominance), not FAA. Table 4.15 gives the exact per-task percentages behind those ranges.

Table 4.15: SHAP diagnostic: mean absolute SHAP importance share (%) by feature family and task

Task	DE	Hjorth	FAA	Connectivity
valence	50.7	21.5	3.7	24.2
arousal	51.8	23.0	3.4	21.8
dominance	47.2	22.9	6.1	23.8
liking	46.8	25.7	4.4	23.1

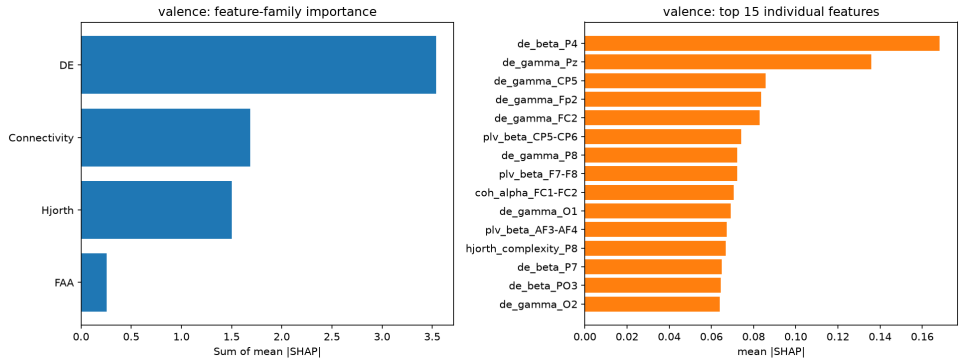


Figure 4.19: SHAP summary: valence (XGBoost, Full feature set, held-out fold).

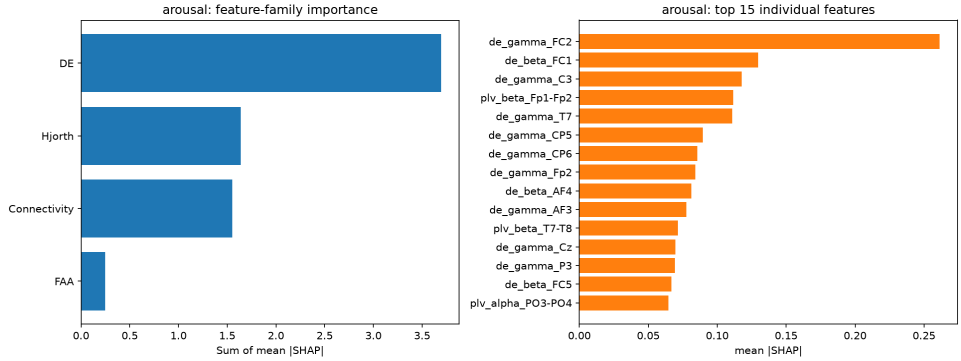


Figure 4.20: SHAP summary: arousal.

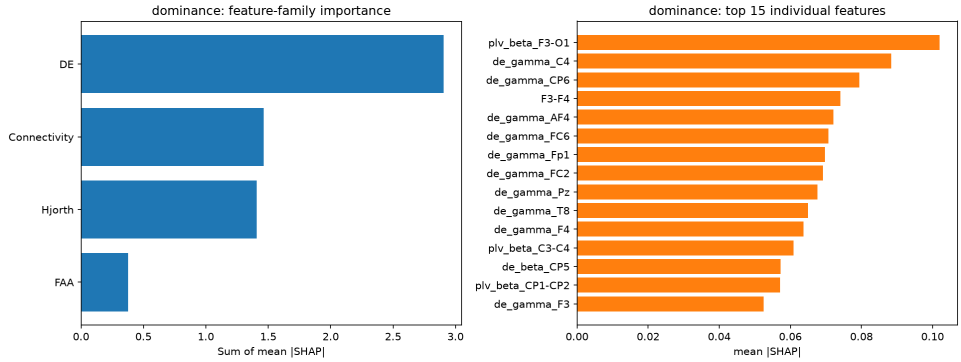


Figure 4.21: SHAP summary: dominance.

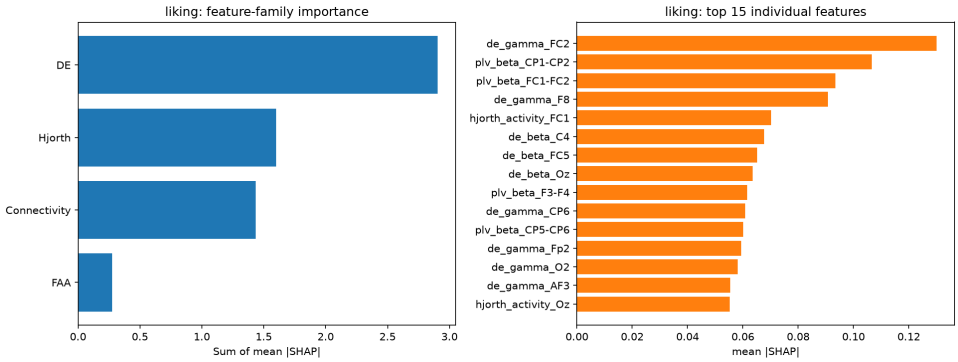


Figure 4.22: SHAP summary: liking.

4.16 Does CORAL stack on top of Euclidean Alignment?

I tested four conditions on the DE-only feature set: no alignment, CORAL alone, EA alone, and EA followed by CORAL on the resulting per-fold split. It does not stack. Mean AUC by condition: EA alone 0.523, EA+CORAL 0.514, no-alignment baseline 0.479, CORAL alone 0.473 (the worst of the four on this feature set). Adding CORAL on top of EA reduces AUC in 10/12 task/model combinations (mean delta −0.0093). One possible reason: EA's per-subject raw-signal whitening already removes most of the between-subject covariance structure CORAL is also designed to correct, so CORAL's additional per-fold alignment acts on residual structure that is mostly noise. The practical takeaway is to use EA alone, not EA+CORAL.

Table 4.16: EA+CORAL stacking: mean ROC-AUC by condition, DE-only feature set, across 4 tasks × 3 models (12 fits)

Condition	Mean ROC-AUC
none (no alignment)	0.479
CORAL alone	0.473
EA alone	0.523
EA + CORAL	0.514

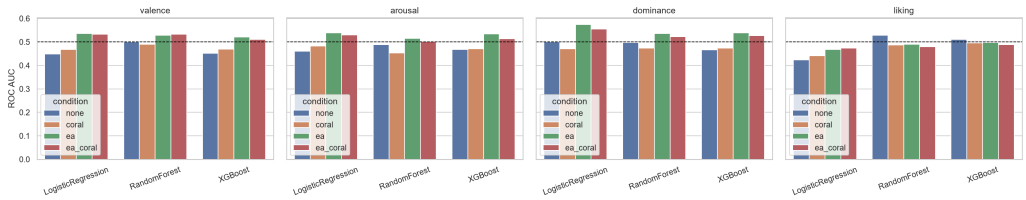


Figure 4.23: EA+CORAL stacking: ROC-AUC by condition, model, and task.

4.17 EEG overview quality-control check

I also built a small-multiples quality-control plot: an 8 × 4 grid, one panel per subject, each showing all 32 channels of that subject's baseline-normalised trial-1 EEG stacked with a fixed vertical offset, in a single neutral ink colour. Per-channel colour-coding was deliberately avoided: with 32 overlapping traces per panel, colour cannot carry per-channel identity meaningfully. This is a visual sanity check, not an analysis. It lets a reader spot subjects with obviously flat, saturated, or anomalously noisy channels before trusting downstream numbers for them.

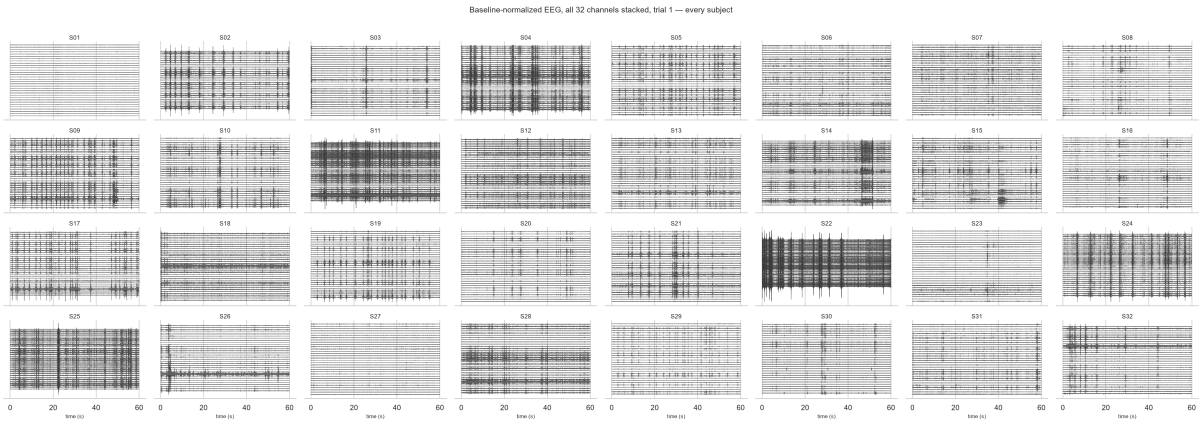


Figure 4.24: EEG overview QC grid: baseline-normalised trial-1 EEG, all 32 channels, one panel per subject (8×4 grid).

Chapter 5

Appendix: Implementation Details and Companion Scripts

This appendix holds the lower-level implementation detail that Chapter 4 omits: function names, cache file names, and the standalone scripts behind the experiments that were too expensive to run inline in the notebook.

5.1 Core pipeline implementation

Two functions handle preprocessing. `load_subject` reads a subject's `.dat` file (a pickled dictionary of EEG data and labels). `baseline_normalize` subtracts the per-channel mean of the pre-trial baseline from the stimulus signal:

```
def load_subject(subject_id):
    fpath = DATA_DIR / f"s{subject_id:02d}.dat"
    with open(fpath, "rb") as f:
        d = pickle.load(f, encoding="latin1")
    return d["data"], d["labels"]

def baseline_normalize(trial_all_channels):
    eeg = trial_all_channels[:N_EEG_CHANNELS]
    baseline = eeg[:, :BASELINE_SAMPLES]
    signal = eeg[:, BASELINE_SAMPLES:BASELINE_SAMPLES + TRIAL_SAMPLES]
    baseline_mean = baseline.mean(axis=-1, keepdims=True)
    return (signal - baseline_mean).astype(np.float64)
```

`compute_de_features` band-pass filters (4th-order Butterworth, zero-phase `sosfiltfilt`) into the four bands, reshapes each band into 15 non-overlapping 512-sample windows, and computes the Differential Entropy of each (channel, band) pair. `make_raw_windows` keeps the raw $(32, 512)$ window and standardises it per channel. A sanity check on subject 1's first trial confirms the expected shapes: `(15, 128)` DE features and `(15, 32, 512)` raw windows.

Configuration constants: `FS = 128`, `BASELINE_SEC = 3`, `TRIAL_SEC = 60`, `WINDOW_SEC = 4`; band edges `theta 4-8, alpha 8-13, beta 13-30, gamma 30-45` Hz; `LABEL_THRESHOLD = 5.0`; `N_EPOCHS = 40`, `BATCH_SIZE = 128`, `PATIENCE = 8` for Track B training. Track B's `subject_independent_split` performs a nested `GroupShuffleSplit` for the 70/15/15 split; `train_torch_model` wraps the `AdamW` + `ReduceLROnPlateau` + mixed-precision training loop described in Section 4.4.

The main windowed dataset is cached to `processed_cache/deap_windows_4s_32subj_v2.npz`; the extended 295-feature set (Section 4.7) to `deap_features_ext_v1.npz`; the 50%-overlap window variant (Section 4.9) to `deap_windows_4s_overlap50_32subj_v1.npz`. Caches are schema-versioned so a stale cache from before dominance/liking were tracked is never silently reused.

5.2 Feature/dataset builders

`dl_models.py` Canonical PyTorch definitions of EEGNet, ShallowConvNet, CNN-LSTM, and TSception, plus the hemisphere-pair channel indices TSception needs. Imported by every `track_b_*.py` script so the architectures are defined exactly once, not re-implemented per script.

`rich_features.py` Implements Hjorth parameters (`compute_hjorth`), frontal/hemispheric alpha asymmetry (`compute_alpha_asymmetry`), and PLV/coherence connectivity (`compute_connectivity`, via a from-scratch Hilbert-transform phase-locking-value computation and a Welch-style magnitude-squared-coherence estimator). Includes a `if__name__=="__main__"` self-test against synthetic data (shape checks, $[0, 1]$-boundedness for PLV/coherence, and an index-math cross-check for FAA) that must pass before the module is trusted for the real build.

`build_extended_features.py` Runs `rich_features.py` (plus DE) over all 32 DEAP subjects, producing `deap_features_ext_v1.npz` (295-dim Full feature set, 19,200 windows). Its DE computation is verified to match the original cache byte-for-byte before the richer features are trusted.

`build_overlap_windows.py` Rebuilds the raw-window Track B dataset with 50% stride overlap (`deap_windows_4s_overlap50_32subj_v1.npz`, 37,120 windows), for Section 4.9.

`build_raw_ratings.py` Extracts the raw continuous 1–9 SAM ratings per window, row-aligned 1:1 with the extended feature cache (verified via an explicit `groups` array equality assertion), for Section 4.12's label-cleanup variants.

5.3 Track A companion scripts

`track_a_ablation.py` Section 4.7: 6 feature-set combinations × 3 models × 4 tasks, `GroupKFold(5)`.

`track_a_repeated_nested_cv.py` Section 4.8: 5×-repeated `GroupKFold` for all 6 models (with a custom subject-relabeling generator, `repeated_group_kfold_splits`, since `sklearn` has no `RepeatedGroupKFold`) plus nested `GroupKFold(3)`/`RandomizedSearc` hyperparameter search for XGBoost and RandomForest.

`track_a_label_cleanup.py` Section 4.12: the three label-binarisation variants (`raw_threshold`, `per_subject_z`, `drop_middle`).

`track_a_calibration_loso.py` Section 4.13: true 32-fold LOSO with an 80/20 per-subject eval/calibration trial split and XGBoost warm-start (`xgb_model=` continuation boosting).

`track_a_significance_test.py` Section 4.14.1: 300-permutation, trial-within-subject label-shuffle null-distribution test, parallelised across CPU cores with `joblib.Parallel`.

`track_a_domain_alignment.py` Section 4.14.2: CORAL alignment (`coral_align`, via `_sym_matrix_power` eigendecomposition).

`track_a_riemannian_alignment.py` Section 4.15.1: Euclidean Alignment (`euclidean_align`), operating on raw windows before DE recomputation.

`track_a_shap_diagnostic.py` Section 4.15.5: `shap.TreeExplainer` on one representative fold per task, aggregated by feature and by family, with bar-chart figures saved per task.

`track_a_ea_coral_stacked.py` Section 4.16: the 4-condition (`none`/`coral`/`ea`/`ea_coral`) stacking test, re-implementing both `euclidean_align` and `coral_align` locally so this script is self-contained.

5.4 Track B companion scripts

`track_b_repeated.py` Section 4.9: 5-seed repeated subject-independent splits, both window variants, all 4 architectures × 4 tasks (160 runs).

`track_b_domain_adversarial.py` Section 4.14.3: `GradReverse` autograd function, `EEGNetDANN` (reuses EEGNet's conv blocks, adds a GRL + domain-classifier branch), and the Ganin & Lempitsky progressive-λ training loop.

`track_b_ssl_pretrain.py` Section 4.15.2: `EEGNetSSL` (shares EEGNet's trunk, regression head for the DE-prediction pretext task), a 15-epoch label-free pretraining stage, then fine-tuning; the training/val/test split is computed once and reused across all four tasks (compute optimisation, not a methodological shortcut, since the split depends only on `groups`/`seed`, not on labels).

`track_b_auc_maximization.py` Section 4.15.3: `auc_margin_loss`, a drop-in loss-function swap with everything else (model, split, seed) held identical to the cross-entropy baseline.

`track_b_test_time_adaptation.py` Section 4.15.4: `reset_bn_for_adaptation` (zeroes BatchNorm running stats, sets momentum to `None` for a true cumulative average) and `adabn_predict_one_subject` (2 label-free adaptation epochs per test subject, under `torch.no_grad()`).

5.5 Cross-corpus scripts

See Section 4.10.2 for `base.py`, `seed_adapter.py`, and `run_cross_corpus.py`.

5.6 Notebook-authoring scripts

A handful of one-off scripts were used to author the notebook's cells during earlier stages of the project. They are not part of the reproducible-results path.

5.7 Script/notebook execution map

There is no automation wiring the scripts and the notebook together – no notebook cell calls a script, and no script calls the notebook. Every Tier-1 companion script above is run manually from a terminal, writes a `.npz`/`.csv`/`.png` file to `processed_cache/` or `results/`, and a separate notebook cell later reads that file back in. `script_notebook_execution_map.txt`, at the repository root, is the hand-verified record of that manual wiring: it groups the builders into a Tier 0 (three scripts that turn the raw DEAP `.dat` files into the `processed_cache/*.npz` caches every later step reads) and a Tier 1 (every companion script, each mapped to the exact notebook section and cell range that consumes its output), notes which caches a script depends on before it will run, and ends with a recommended clean-checkout run order. It also flags a concrete rough edge: three of the Tier-0 builders (`build_extended_features.py`, `build_raw_ratings.py`, `build_overlap_windows.py`) still hardcode an old `D:/EEG/...` path for `DATA_DIR`/`CACHE_DIR` from an earlier stage of the project, unlike `track_a_significance_test.py`, which was already migrated to a relative, repository-root-based path; those three need that path fixed (or need to be run from the original location) before a clean checkout can reproduce their caches. The map is generated by inspecting the current repository state rather than hand-maintained prose, so it is the more reliable of the two if a script or notebook cell is ever added, removed, or renamed after this report is written.

Chapter 6

Literature Review and Bibliography

6.1 Corpus

`papers_EEG/` contains 32 papers I reviewed for this project, each renamed to a descriptive title. Two plain-text files at the repository root track this corpus outside of LaTeX: `citations.txt` is the flat citation list (the source for the `bibitem` entries in this chapter's bibliography), and `literature_log.txt` is a per-paper log of what each paper covers and, where applicable, which section of this report it fed into; it was reconstructed in one pass late in the project by reading all 32 papers directly (Chapter 12 discusses why). Table 6.1 lists the 13 with an explicit, documented role in an experiment: named as the specific motivation behind a specific intervention.

Table 6.1: Papers with an explicit, documented contribution to this project

Paper	Used in	Why
Khan et al. (2025)	Section 4.14.1	Documents that DEAP pipelines can look deceptively strong under leaky evaluation, even on a label-randomised control. This motivated building a permutation-null significance test for my own `GroupKFold` protocol.
Sarafraz et al. (2022)	Sections 4.14.2, 4.14.3, 4.15.1	Surveys CORAL and Euclidean Alignment as standard, cheap, unsupervised domain-adaptation remedies for distribution shift, and is one of several sources converging on adversarial subject-invariance (DANN).
Ye et al. (2023)	Sections 4.14.2, 4.15.1	Reports plain CORAL reaching roughly 65–70% under LOSO on SEED, where untransformed features stay under 60%: the empirical precedent that motivated testing CORAL and EA here, though my own CORAL result did not replicate that scale of gain.
Zhang et al. (2022)	Section 4.14.3	One of several independently-converging sources recommending a gradient-reversal-layer domain discriminator for subject-invariant EEG representation learning.

Paper	Used in	Why
Li et al. (2023)	Section 4.14.3	Same DANN motivation as above.
Liu et al. (2024a)	Section 4.14.3	Same DANN motivation as above.
Liu et al. (2025a)	Section 4.14.3	Same DANN motivation as above.
Sreehari et al. (2026)	Section 4.14.3	Names over-aggressive alignment collapsing class structure as a documented failure pattern in unsupervised domain adaptation, matched by my own DANN arousal/dominance collapse and used to interpret that result.
Wang et al. (2024)	Section 4.15.2	The one paper in the review that evaluated LOSO on DEAP itself while beating domain-adaptation baselines: the strongest outside evidence that a pretraining stage, rather than a training-time loss or architecture change, could move subject-independent AUC.
Weng et al. (2024)	Section 4.15.2	Corroborating survey context for the SSL-pretraining design.
Xiao (2024)	Section 4.15.3	The most directly on-metric deferred idea: my headline metric is ROC-AUC, so a loss that directly targets AUC, rather than per-sample cross-entropy, is a natural, cheap intervention to test.
Imtiaz & Khan (2025)	Section 4.15.4	Surveys the source-free UDA family, of which AdaBN is the cheapest, most-established member, as a natural target-aware follow-up to DANN.
Mouazen et al. (2025)	Section 4.15.5	Motivated logging the SHAP interpretability diagnostic as a cheap check on whether near-chance models lean on plausible or noise-like features.

6.2 Papers reviewed but not tied to a specific experiment

I list these 19 papers separately, rather than silently dropping them from the corpus once they did not produce a script, for the same transparency reason the retracted permutation-test claim in Section 4.14.1 is kept rather than deleted: a literature review that only shows the sources that paid off gives a misleading picture of how much reading actually went into the design decisions above, and hides the fact that several plausible directions (GAN/diffusion-based data augmentation, graph-neural-network and transformer architectures, contrastive self-supervised pretraining beyond the one variant I did implement) were considered and set aside rather than never considered at all. The remaining 19 papers in the corpus were part of the same review pass (EEG-emotion surveys, domain-adaptation and generalisation methods, contrastive/self-supervised pretraining, GAN/diffusion augmentation, and DEAP-specific methodology critiques) but did not drive a specific implemented experiment, generally because they targeted a different dataset,

required an architecture change larger than a retrofit onto the ten base models already built, or converged on a recommendation (e.g. domain adaptation, self-supervised pretraining) that an already-cited paper in Table 6.1 had independently motivated with a clearer, more directly applicable empirical precedent: Liu et al. (2021), Qiao et al. (2022), Menezes et al. (2017), Xu et al. (2022), Guo et al. (2023), Kan et al. (2022), Siddhad et al. (2025), Liu et al. (2024b), Liu et al. (2025b), Wang et al. (2025), Gong et al. (2025), Dolgopolyi & Chatzipanagiotou (2025), Purwar et al. (2026), Kalashami et al. (2022), Pan & Zheng (2021), Liu et al. (2024c), Hamzah & Abdalla (2024), Azar et al. (2024), and Karthiga et al. (2024). They are listed here as background and related work rather than trimmed from the corpus, since they informed my sense of the field even where they did not map to a specific script.

Chapter 7

Consolidated Results

Chapter 4 presents each experiment's numbers embedded in its own methods and discussion, which is the right place to read any one result in context but makes it hard to see the project's overall shape at a glance: whether the interventions cluster into a few that worked and many that did not, or whether the effects are uniformly small throughout. Table 7.1 consolidates the headline AUC numbers across every experiment in the notebook, in chronological/section order, specifically to make that shape visible in one place. All numbers are mean ROC-AUC unless noted; "Δ" is the change versus that experiment's own stated control condition. Reading down the Verdict column, only two of the fourteen listed interventions earn an unqualified positive verdict (Euclidean Alignment and, on the one task it targets, the AUC-margin loss); the great majority land on "no stable effect," "inconsistent, not real gain," or an explicit negative (DANN's arousal/dominance collapse, EA+CORAL underperforming EA alone). That imbalance, more than any single row, is the table's real content: it is a compact visual record of the same negative-result pattern that Section 4.11 and Chapter 8 argue for in prose, now readable as fourteen data points rather than as an argument that has to be taken on trust.

Table 7.1: Master results table

Experiment (section)	Best/observed AUC	Δ vs. control	Verdict
Track A best model (§5)	0.501–0.534	n/a	near chance
Track B best model (§6)	0.475–0.607	n/a	near chance, mixed by task
Feature-set ablation (§9)	0.478–0.494 (means)	$\leq$ 0.016 across sets	no feature set helps
Repeated GroupKFold (§10)	15/24 CIs contain 0.5	n/a	mostly chance-consistent
Nested CV (§10)	0.454–0.524	$\approx$ 0 vs. fixed HP	tuning is not the bottleneck
Window overlap (§11)	0.483 vs. 0.498	+0.015	inconsistent, not real gain
Cross-corpus DEAP baseline (§12)	0.452	n/a	weakest of the three settings
Cross-corpus zero-shot (§12)	0.517	+0.065 vs. DEAP baseline	modest, above chance
Cross-corpus SEED-IV LOSO (§12)	0.673	+0.221 vs. DEAP baseline	moderate signal

Experiment (section)	Best/observed AUC	Δ vs. control	Verdict
Label cleanup, per_subject_z (§14)	0.514 (mean)	+0.031 vs. raw	modest, confounded by rebalancing
Calibration, arousal (§15)	0.577	+0.071 vs. zero-shot	modest personalisation effect
Permutation test, dominance (§16.1)	0.561	$z = +2.18$, $p = 0.0199$	significant but small, not Bonferroni-robust
Permutation test, others (§16.1)	0.437–0.458	$p = 0.877$–0.997	not distinguishable from null
CORAL (§16.2)	mean $\Delta + 0.0059$	8/12 up, 4/12 down	no stable effect
DANN, valence/liking (§16.3)	+0.039/+0.050	n/a	small gains
DANN, arousal/dominance (§16.3)	−0.216/−0.149	n/a	**collapse**
Euclidean Alignment (§17.1)	mean $\Delta + 0.044$	10/12 up	**strongest effect overall**
SSL pretraining (§17.2)	+0.074/+0.100 (val/lik)	−0.082/−0.053 (aro/dom)	bimodal task split
AUC-margin loss (§17.3)	+0.083 (valence)	mixed elsewhere	mechanistically understood win
AdaBN (§17.4)	+0.082 (liking)	−0.101/−0.063 (aro/dom)	net negative
SHAP diagnostic (§17.5)	n/a	n/a	diffuse importance, FAA weakest
EA+CORAL stacking (§18)	EA 0.523 vs. EA+CORAL 0.514	−0.0093	**does not stack; use EA alone**

7.1 Figures

All 24 figures in `results/figures/` are embedded in place in Chapter 4, alongside the section that produced them (see List of Figures, p. 12, for the full index). The most important for a final report remain `class_balance.png` (Fig. 4.1, §4), `track_a_accuracy.png`/`combined_accuracy_comparison.png` (Figs. 4.2/4.4, §5–7), `track_a_repeated_cv_auc_ci.png` (Fig. 4.7, §10), `track_a_riemannian_alignment_auc.png` (Fig. 4.15, §17.1, the strongest result), `track_a_ea_coral_stacked_auc.png` (Fig. 4.23, §18), and `cross_corpus_deap_seediv.png` (Fig. 4.9, §12).

Chapter 8

Model-by-Model Analysis

Chapters 4 and 7 report what each of the ten models scored. This chapter looks at why, for each model, at the level of inductive bias, capacity, and training procedure. These are my working hypotheses for each score, not confirmed mechanisms: none of them were tested directly against an alternative explanation, and most of the differences between models are small relative to fold-to-fold noise. Where an effect is large enough to be worth a fuller explanation, I say so and note it as a hypothesis; where a difference is comparable to normal fold-to-fold variation, I say that instead of inventing a story for it.

8.1 A shared ceiling: what every model is fighting against

Five properties of the DEAP setup apply to all ten models and are worth holding in mind when reading any individual score:

- Class imbalance is task-dependent and interacts with accuracy. Valence (21.0% high) and arousal (23.3% high) reward any model that leans on the majority class; dominance (60.9% high) and liking (66.5% high) are close enough to balanced that accuracy and AUC tend to agree more (Section 4.2). A model's accuracy on valence/arousal is only informative once checked against its AUC: high accuracy with AUC near 0.5 is the majority-class-collapse signature discussed in Section 4.6.

- Only 32 subjects supply the grouping variable. Under `GroupKFold(5)`/`GroupShuffleSplit`, each held-out fold is about 6–7 entire subjects the model has never seen. With this few groups, between-subject variance in raw EEG amplitude, electrode impedance, and baseline arousal state is a large fraction of total variance, large enough that some fold-to-fold swings are driven more by which subjects landed in which fold than by any model property. This is a background noise floor that no architecture change removes, and it is a large part of why Section 4.8's repeated 5×5 GroupKFold shows 15/24 95% confidence intervals still spanning chance.

- The labels are self-report, thresholded 1–9 ratings, not a physiological ground truth. Section 4.12's label-cleanup ablation shows that even changing how the same raw ratings are binarised moves mean AUC by at most +0.031, so no model, however expressive, can be expected to find much more signal than the label construction itself contains.

- Track A and Track B are evaluated under different variance regimes. Track A's numbers are 5-fold `GroupKFold` means; Track B's headline numbers (Table 4.2) are single-split point estimates, so an individual Track B score is noisier and more fold-luck-dependent than a Track A one (Section 4.9's repeated splits give mean AUC 0.483 non-overlap, consistent with the single-split numbers here).

- All 32 DEAP subjects watched the same 40 stimulus videos. Any component of arousal or valence that is driven by the stimulus rather than the person is, in effect, partially shared across subjects even under a leakage-free `GroupKFold` split: no subject's windows leak, but the video each window comes from is seen by the model during training via other subjects' reactions to it. This may be part of why arousal tends to score a little better than the other three tasks for several models, though all four stay close to chance overall, without implying any actual data leakage.

8.2 Track A: classical models on DE features

All six models share the same 128-dim Differential Entropy input, the same standard-scaling preprocessing, and the same 5-fold `GroupKFold` protocol (Section 4.3), so differences between them are attributable to model family, not to differences in the data they see (except SVM-RBF's 8000-sample training cap). Table 8.1 collects every model's mean ROC-AUC on every task.

Table 8.1: Track A – every classical model's mean ROC-AUC, all four tasks (5-fold GroupKFold)

Model	Valence	Arousal	Dominance	Liking
LogisticRegression	0.449	0.461	0.500	0.424
kNN ($k = 15$)	0.465	0.495	0.501	0.531
RandomForest (300 trees)	0.501	0.489	0.497	0.528
SVM-RBF ($C = 10$)	0.500	0.450	0.513	0.534
XGBoost (depth 6)	0.455	0.483	0.473	0.505
MLP (128,64)	0.457	0.439	0.493	0.488

8.2.1 Logistic Regression

The only strictly linear model in the comparison, and the one I reused as the fast proxy in the permutation significance test (Section 4.14.1) and inside CORAL alignment (Section 4.14.2). Its AUC sits at or below chance on three of four tasks (0.449 valence, 0.461 arousal, 0.424 liking, the single worst score of any model on any task in Track A) and only reaches 0.500 on dominance. One possible reason: the SHAP diagnostic (Section 4.15.5) finds class-relevant structure diffuse across dozens of features with no single dominant direction, and a linear hyperplane has limited ability to combine that kind of structure into a useful score. Liking's especially poor score, meaningfully below 0.5, is consistent with this model's role as the null-hypothesis proxy in Section 4.14.1: it can be pushed to the wrong side of chance by fold-to-fold subject swapping. It is also the fastest model in Track A.

8.2.2 k-Nearest Neighbours

The only non-parametric model in the comparison: it classifies by majority vote among each test point's 15 nearest neighbours in 128-dimensional space. It is never the single best model in Table 8.1, but it is the best of the six on arousal (0.495) and the second-best on liking (0.531, behind SVM-RBF). One possible reason is that kNN has no training-time optimisation that can overfit a training fold's particular mix of subjects, since its only free parameter is k, fixed in advance. It does not lead on valence (0.465) or dominance (0.501); the differences between kNN and the other classical models on those two tasks are small enough to be within normal fold-to-fold variation rather than a model-specific effect worth explaining further.

8.2.3 Random Forest

With scikit-learn's default unlimited tree depth, Random Forest is the single best model by AUC on valence (0.501) and by a wide margin the best by accuracy on every task (0.779 valence, 0.767 arousal, 0.577 dominance, 0.643 liking). This accuracy/AUC split is one of the clearest illustrations in this report of the majority-class-collapse pattern discussed in Section 4.6: bagged trees can find splits that predict the majority class on valence/arousal with high accuracy without needing the minority class to be separable. Averaging over 300 trees also gives it the tightest accuracy variance in Track A, which makes it look stable, but stability of an accuracy number built partly on base-rate exploitation is not the same as stability of a real discriminative signal; its AUC (0.489–0.501) is unremarkable once that base rate is accounted for.

8.2.4 Support Vector Machine (RBF kernel)

Trained on at most 8000 randomly subsampled windows per fold, SVM-RBF is the best model by AUC on two of four tasks: dominance (0.513) and liking (0.534, the single highest AUC anywhere in Track A). It is also the slowest model by 1–2 orders of magnitude, and has the lowest accuracy of any model on valence (0.676) and arousal (0.691): it is the one classical model that consistently avoids leaning on the majority class, plausibly because its internal probability calibration smooths output scores away from the overconfident majority-class predictions that inflate other models' accuracy without inflating their AUC. The 8000-sample cap is worth flagging as a real confound: on valence and arousal, where the full training fold has over 15,000 windows, SVM-RBF sees under 60% of the available data, which may cost it AUC on exactly the two tasks where it under-performs kNN and Random Forest.

8.2.5 XGBoost

XGBoost is never the best Track A model on any task (0.455 valence, 0.483 arousal, 0.473 dominance, 0.505 liking), despite being the model I chose to represent Track A in the feature-set ablation (Section 4.7) and the SHAP diagnostic (Section 4.15.5). Gradient boosting fits trees sequentially to residual error, which can make it more prone than bagging to chasing patterns specific to a training fold's particular subject mix. Consistent with this, nested hyperparameter search (Section 4.8) barely moves its AUC (dominance: 0.494 tuned vs. 0.473 fixed hyperparameters), so tuning does not appear to be the main bottleneck. XGBoost is the cheapest model per fit after kNN, which is why I used it for the many companion scripts that run hundreds of fits (ablation, label-cleanup, calibration LOSO).

8.2.6 MLP (shallow feed-forward network)

The only model in Track A with a trainable nonlinear architecture, and it is nonetheless never the best model on any task and is the single worst model on arousal (0.439, below even Logistic Regression's 0.461). This is a meaningful negative result on its own: if two extra hidden layers on the same 128 DE features cannot beat a linear model, the ceiling is unlikely to be that classical features need a nonlinear decision function, which lines up with Section 4.7's finding that richer feature sets do not move AUC either. Its accuracy variance on valence is the widest in Track A, consistent with early stopping shrinking its effective training data.

8.3 Track B: deep models on raw EEG windows

All four architectures share the same raw, baseline-normalized 32-channel × 512-sample (4 s) input, the same `AdamW` + `ReduceLROnPlateau` + early-stopping (patience 8) training loop, and the same single 70/15/15 subject-independent split (Section 4.4), so differences are attributable to architecture and parameter count. Table 8.2 collects every model's ROC-AUC on every task.

Table 8.2: Track B – every deep model's ROC-AUC, all four tasks (single subject-independent split)

Model	Params	Valence	Arousal	Dominance	Liking
EEGNet	2,130	0.561	0.607	**0.288**	0.423
ShallowConvNet	54,562	0.468	0.557	0.408	0.439
CNN-LSTM	82,466	0.558	0.605	0.452	0.497
TSception	8,765	0.498	0.483	0.475	0.472

8.3.1 EEGNet

EEGNet (2130 parameters, the smallest architecture by a wide margin) posts the best or tied-best AUC on valence (0.561) and arousal (0.607), and the single worst result in the entire project, 0.288 AUC on dominance (Section 4.4): meaningfully below chance rather than merely noisy, and large enough to be worth a fuller explanation. One hypothesis: EEGNet's depthwise spatial convolution learns one fixed spatial weighting across all 32 electrodes per filter, shared identically across every subject. This can be a useful inductive bias when the target correlates with a consistent scalp topography shared across subjects, plausible for arousal and valence, but a liability when no such topography exists. With only 2130 parameters total, EEGNet has little spare capacity to hedge: if its one learned spatial filter for a given task happens to align with the training folds' subject-specific noise rather than a pattern common across subjects, there is no redundant filter to fall back on, and a single-split evaluation with no error bars (Section 8.1) can land on a filter whose sign is inverted relative to the held-out subjects, producing an AUC below 0.5 rather than merely near it. Dominance has the weakest, most diffuse connection to any engineered signal elsewhere in this report (the SHAP diagnostic, Section 4.15.5, finds no dominant feature for it either), which may make it the task most likely to expose this failure mode. This same fragility, if the hypothesis is right, would explain why the arousal/dominance instability recurs across every intervention built on EEGNet in Sections 4.14 and 4.15 (DANN, SSL pretraining, AUC-margin loss, AdaBN). I have not tested this against a second architecture, so it remains a hypothesis rather than a confirmed cause.

8.3.2 ShallowConvNet

The largest convolutional architecture (54,562 parameters) is never the best model on any task and is the most uniformly mediocre: 0.468 valence, 0.557 arousal, 0.408 dominance, 0.439 liking. No catastrophic failure like EEGNet's, but no real win either. Its square-nonlinearity-then-log pooling stage is modelled on FBCSP-style log-band-power features, making it the Track B architecture conceptually closest to Track A's DE pipeline, except it must learn its own baseline normalisation from scratch. With more parameters than EEGNet but the same data to train on, it may have more capacity to fit training-fold-specific signal scales that do not transfer, which would be consistent with its accuracy sitting close to the majority-class rate on valence while its AUC lags EEGNet's.

8.3.3 CNN-LSTM

The largest architecture overall (82,466 parameters) is the best or near-best model on three of four tasks: tied for best on valence (0.558 vs. EEGNet's 0.561), second-best on arousal (0.605), and outright best on liking (0.497), while remaining weak on dominance (0.452). Its CNN front end collapses the channel dimension via a spatial convolution, then a bidirectional LSTM models the resulting temporal sequence, giving it the ability to weight different parts of the 4-second window differently rather than pooling it into one fixed-size descriptor. One possibility is that

arousal and liking are the two tasks where within-window temporal dynamics carry information beyond a static band-power summary, which would explain why the extra temporal modelling helps there and not on dominance. CNN-LSTM and EEGNet converge to numerically identical 83.5% accuracy on valence via different mechanisms, which is independent evidence that valence's 79% skew, more than either architecture's inductive bias, sets the accuracy number on that task.

8.3.4 TSception

TSception (8765 parameters), built around a hemisphere-asymmetry branch modelling the frontal-alpha-asymmetry valence hypothesis, is the worst of the four models on the task its architecture targets: valence, 0.498 AUC, versus EEGNet/CNN-LSTM's 0.56-plus, while being the best of the four on dominance (0.475, still below chance but the least-bad Track B result on that task). This lines up with the SHAP diagnostic (Section 4.15.5), which independently found frontal/hemispheric alpha asymmetry to carry the smallest importance share of any feature family on every task: two different modelling paradigms agreeing that this specific hypothesis does not help here is suggestive, though neither result alone proves the other. TSception's relative strength on dominance may owe more to its multi-scale temporal branches acting as an implicit ensemble within a small network than to any asymmetry-specific advantage.

8.4 Cross-model synthesis: why ten different algorithms converge on the same ceiling

The single most informative fact in this chapter is not any individual model's score, but that ten models spanning five different inductive biases (a linear hyperplane, a lazy distance-based vote, two tree ensembles built by different objectives, a kernel method, a generic nonlinear function approximator, and four deep architectures with different spatial/temporal priors) all land in the same 0.29–0.61 AUC band on the same four tasks (Section 4.6). If the ceiling were a property of any one model family, swapping model families should have produced at least one clear escape from the band; it does not. This is the model-level version of the argument Section 4.11 makes at the experiment level: a ceiling that survives across bagging, boosting, kernels, lazy learning, shallow MLPs, and four structurally distinct deep architectures is more consistent with a property of the labels and subject shift than with any one model's expressiveness. That reading is independently supported by the 300-permutation significance test (Section 4.14.1) and, cautiously, by the SEED-IV cross-corpus result (Section 4.10): the identical DE+XGBoost pipeline reaches a moderate 0.673 AUC on a different corpus, which is consistent with the ceiling being at least partly DEAP-specific, though that comparison rests on a single model and a single other dataset.

Two secondary patterns recur across both tracks and are visible only once every model is compared side by side:

- Accuracy and AUC decouple in the direction imbalance predicts, for every high-capacity model on every imbalanced task. Random Forest, XGBoost, EEGNet, and ShallowConvNet all post their highest accuracy, and their least impressive AUC relative to their own best task, on valence/arousal; the lower-capacity models (Logistic Regression, SVM-RBF, kNN, TSception) show a smaller accuracy/AUC gap on those same tasks.

- Every architecture or model built around a specific, named neurophysiological hypothesis under-performs models with no such hypothesis, on the exact task the hypothesis targets: TSception's asymmetry branch underperforms on valence, and the SHAP diagnostic's FAA features carry the least importance of any family on every task (Section 4.15.5). Two independent modelling paradigms agreeing that hand-encoding this hypothesis does not help is a stronger signal than either result alone, though it is still only two data points.

8.5 Beyond the base ten: why the model variants and extensions scored what they scored

Sections 8.2–8.3 explain the ten base model/task results. Every further experiment in the notebook (Sections 4.7–4.16) is not a new model, but one of the ten base models – almost always Logistic Regression, RandomForest, and/or XGBoost for Track A, or EEGNet for Track B – retrained under a modified feature set, label definition, alignment transform, loss, or evaluation protocol. This section walks through each such variant and explains its score *in terms of the base model's own mechanism*, as diagnosed above, rather than repeating the numbers in isolation. Table 8.3 indexes which base model(s) each variant reuses.

Table 8.3: Every model variant/extension beyond the base ten, and which base model(s) it reuses

Variant	Base model(s) reused	Section
Richer feature sets (Hjorth/FAA/connectivity)	LR, RandomForest, XGBoost	§7
Repeated & nested GroupKFold	RandomForest, XGBoost	§8
50% overlapping windows	EEGNet, ShallowConvNet, CNN-LSTM, TSception	§9
Label-cleanup ablation	LR, RandomForest, XGBoost	§12
Per-subject calibration (LOSO)	XGBoost	§13
Permutation significance test	LogisticRegression	§14.1
CORAL feature alignment	LR, RandomForest, XGBoost	§14.2
Subject-adversarial (DANN)	EEGNet	§14.3
Euclidean Alignment	LR, RandomForest, XGBoost	§15.1
SSL (TFR-style) pretraining	EEGNet	§15.2
AUC-margin training loss	EEGNet	§15.3
AdaBN test-time adaptation	EEGNet	§15.4
SHAP interpretability diagnostic	XGBoost	§15.5
EA+CORAL stacking	LR, RandomForest, XGBoost	§16

8.5.1 Richer feature sets, repeated/nested CV, window overlap, label cleanup (Sections 4.7, 4.8, 4.9, 4.12)

These four extensions all move mean AUC by small amounts that sit within, or close to, normal fold-to-fold variation, so I do not think they warrant individual mechanistic stories. Richer feature sets move Logistic Regression, Random Forest, and XGBoost's AUC by no more than 0.016 across six feature-set combinations (0.478–0.494); nested hyperparameter search moves XGBoost's dominance AUC from 0.473 (fixed) to 0.494 (tuned), a small +0.021 change; 50% window overlap moves Track B's mean AUC from 0.483 to 0.498; and label cleanup's best variant, per-subject z-scoring, reaches 0.514 versus a 0.483 raw-threshold baseline. In each case the direction is either inconsistent across models/tasks or small enough that I would not treat

it as a real effect without a larger, dedicated test. One possible reason all four fall flat: per Section 4.15.5's SHAP diagnostic, class-relevant structure is diffuse across many features with no dominant direction, so neither adding features, tuning trees, adding correlated windows, nor relabelling changes that underlying picture much.

8.5.2 Per-subject calibration (Section 4.13)

XGBoost run under full 32-fold Leave-One-Subject-Out, zero-shot then warm-started on 8 of the held-out subject's own trials, is one of the larger effects in this report: zero-shot AUC sits around 0.50–0.51, and calibrated AUC rises to 0.577 (arousal), 0.519 (dominance), 0.539 (liking), 0.537 (valence). One possible reason is that warm-starting extra trees on the target subject's own data lets the model pick up subject-specific structure that a fully subject-independent protocol is not permitted to use. This suggests part of the ceiling may be learnable subject-specific structure rather than pure noise, though it is one model and one calibration scheme.

8.5.3 Permutation significance test (Section 4.14.1)

Logistic Regression, the fastest and most conservative choice available in Track A, was used as the proxy for 300 trial-within-subject label-shuffled reruns per task. Under the corrected, trial-respecting null, valence, arousal, and liking's observed AUC is not distinguishable from the null ($p = 0.877$–0.997); dominance is the only task whose observed AUC clears the null, and only barely ($p = 0.0199$, not significant after a Bonferroni correction for four tasks). See Section 4.14.1 for the earlier, buggy window-level-shuffle version of this test and why it overstated every effect.

8.5.4 CORAL feature alignment (Section 4.14.2)

Mean AUC delta from CORAL is +0.0059 across Logistic Regression, Random Forest, and XGBoost (8/12 up, 4/12 down): small and inconsistent in sign, consistent with normal fold-to-fold noise rather than a real effect. CORAL re-colours feature covariance per fold; one possible reason it does little here is that the class-relevant signal in these features is already weak and diffuse (Section 4.15.5), so matching second-moment statistics between train and test does not create a discriminative direction that was not there before.

8.5.5 Subject-adversarial (DANN) training (Section 4.14.3)

Adding a gradient-reversal subject classifier to EEGNet is one of the larger effects in this report. Valence/liking improved (+0.039/+0.050); arousal/dominance collapsed (−0.216/−0.149, precision and recall pinned near 0.50). One hypothesis, following from Section 8.3.1's account of EEGNet's single-spatial-filter fragility: forcing the encoder to also make its representation uninformative about subject identity perturbs that same filter, and where the baseline was already close to a sign-inversion failure (dominance, arousal), that push may be enough to tip it into full class collapse. I have not tested this against a calmer training schedule, which the notebook itself flags as the natural next step (Chapter 11).

8.5.6 Euclidean Alignment (Section 4.15.1)

This is the strongest and most consistent effect in the whole project: mean AUC delta +0.044 across Logistic Regression, Random Forest, and XGBoost (10/12 combinations improved, sign consistent within three of the four tasks). EA whitens each subject's raw channel covariance before DE features are even computed, upstream of where CORAL operates. One possible reason it outperforms CORAL (Section 8.5.4) is that intervening on raw, per-subject covariance removes a subject-specific confound before it can contaminate any downstream feature, rather

than trying to correct for it afterward on a per-fold basis. This is a hypothesis consistent with the pattern observed across all three model families, not a mechanism I verified directly.

8.5.7 SSL pretraining, AUC-margin loss, AdaBN (Sections 4.15.2, 4.15.3, 4.15.4)

SSL pretraining on EEGNet shows the same bimodal task split as DANN (valence +0.074, liking +0.100; arousal −0.082, dominance −0.053), plausibly for a related reason: pretraining sets the initial weights of the same fragile spatial filter discussed in Section 8.3.1, which may help or hurt depending on how well the pretext task's structure lines up with each task's true signal. AdaBN's task pattern (liking +0.082; arousal −0.101, dominance −0.063; valence flat) is small enough on three of four tasks to be consistent with noise, and only adjusts normalization statistics rather than the filter weights themselves, so I would not expect it to fix a filter-direction problem even if that diagnosis is right. AUC-margin loss is the one intervention here with a mechanism I am fairly confident in: it improves valence most (+0.083), and valence is the one task independently diagnosed (Section 8.1) as a majority-class collapse (83.5% accuracy at 0.474 AUC), which an AUC-insensitive-to-imbalance loss directly targets. Its near-zero deltas on the other three tasks, which are not majority-class-collapse tasks to begin with, are consistent with that same explanation.

8.5.8 SHAP interpretability diagnostic (Section 4.15.5)

XGBoost was used for SHAP analysis because `TreeExplainer` is exact and cheap only for tree ensembles. Its finding, diffuse importance across feature families with no single feature above a few percent and frontal alpha asymmetry the weakest family on every task, is consistent with the pattern invoked elsewhere in this chapter (Logistic Regression's flatness, XGBoost's resistance to richer features, TSception's underperformance on valence): several independent readings pointing the same way, though each individually is circumstantial.

8.5.9 EA+CORAL stacking (Section 4.16)

Stacking CORAL on top of EA-whitened features (0.514) underperforms EA alone (0.523), and CORAL alone is the worst of the four conditions tested (0.473 versus a 0.479 no-alignment baseline). One possible reason is that EA already removes most of the subject-specific covariance structure CORAL also targets, so CORAL's remaining effect is closer to noise from matching a per-fold rather than per-subject covariance estimate. The practical takeaway is to use EA alone.

Chapter 9

What Was Accomplished

This chapter, and the three that follow it, exist because a report that only narrates results chronologically, as Chapter 4 does, leaves the reader to reconstruct for themselves what the project actually added up to. Listed together here, deliberately separated from the negative results and the open items that come after, are the concrete, positive contributions of this work: what was built, what was formally established, and what would need to be true for someone else to trust and reuse it.

- A subject-independent, leakage-audited comparison of 6 classical models and 4 deep architectures across all 4 DEAP label dimensions (10 models $\times$ 4 tasks = 40 head-to-head results).
- A negative-result campaign covering 8 or more independent hypotheses for the near-chance ceiling (richer features, hyperparameter tuning, more CV rigor, more data via window overlap, better label definitions, feature alignment via CORAL, subject-adversarial training, and, with a modest exception, per-subject calibration).
- A formal statistical treatment of the central claim: a 300-permutation, trial-within-subject significance test showing that observed AUC for three of four tasks is not distinguishable from a properly-built permutation null, and that dominance's apparent exception is small and does not survive a multiple-comparisons correction.
- Five further literature-motivated techniques implemented end to end (Euclidean Alignment, SSL pretraining, AUC-margin loss, AdaBN, SHAP), producing the strongest result in the project (EA, +0.044 mean AUC) and one well-understood win (AUC-margin loss on valence).
- A cross-corpus validation against SEED-IV, including a from-scratch dataset adapter that handles a channel-order filename mismatch and missing label files, and a result suggesting the near-chance ceiling may be specific to DEAP rather than a general property of subject-independent EEG decoding.
- Reproducibility: the entire notebook runs end to end with zero errors, and every figure and table referenced in it is backed by a file on disk.
- A visual QC pass over the raw data itself, the per-subject EEG overview grid.

Chapter 10

What Was Not Done

Every item below was considered and consciously left out, for a stated reason (data access, compute cost, or scope), rather than overlooked; I list them here, separately from Chapter 11's process lessons, because a reader deciding whether to trust or extend this work needs to know exactly where its boundary sits, not just what is inside it. Several of these directly bound how strongly the conclusions in Chapter 13 can be stated: the cross-corpus claim in Section 4.10, for instance, would be considerably stronger if the two gated datasets below had also been tested.

- 19 of the 32 papers in the corpus (Section 6.2) have no code-level contribution, only a reviewed topical summary.
- DREAMER and AMIGOS cross-corpus validation. Both remain gated behind signed data-use agreements; only their adapter interface was scoped, not implemented.
- Full Leave-One-Subject-Out for Track B. Every deep-learning experiment uses a single or repeated 70/15/15 split; a true 32-fold LOSO, as was done for Track A/XGBoost calibration, was never run for any deep architecture, for cost reasons (roughly 10–30× slower).
- Continuous regression on the 1–9 ratings. Every experiment stays within binarised classification; RMSE/CCC regression on the raw scale was never attempted.
- Data augmentation for Track B (channel dropout, time-warping, mixup). Only 50% window overlap was tried, which is augmentation-adjacent rather than equivalent.
- Grad-CAM or saliency-map comparison for Track B, to compare against the Track A SHAP diagnostic's findings. Never implemented.
- A calmer DANN λ-schedule re-run. The standard Ganin & Lempitsky schedule is a likely cause of the arousal/dominance collapse; a lower max lambda, slower ramp, or gradient-clipping the domain branch is the natural next debugging step, not run.
- Extending DANN to the other three Track B architectures. Deferred pending the λ-schedule fix above.
- GPU acceleration for XGBoost in most Track A companion scripts. Several call `tree_method="hist"` without `device="cuda"`, so they ran on CPU despite the project having a capable GPU, while the main notebook's own Track A cell requests the GPU explicitly.
- DEAP session/channel-order variant robustness. The pipeline assumes the 32-channel `data_preprocessed_python` release; no attempt was made to test against DEAP's alternative distributions (e.g. the raw 48-channel release).

Chapter 11

What Could Have Been Done Differently

Unlike Chapter 10, which lists work that was never attempted, every item here is a process choice I did make, and would make differently with the benefit of hindsight, even though each one seemed reasonable at the time it was made. I include this chapter because several of these choices cost real time later in the project (the literature-log reconstruction in Chapter 12 being the clearest example), and because a future continuation of this work, by me or by someone else, should not have to rediscover the same friction points from scratch.

- Write the literature log as papers were processed, with a running, per-paper note at ingestion time, rather than as a single pass at the end.
- Standardise the XGBoost `device=` parameter across all scripts from the start, via a single shared helper module, rather than each Track A script redefining its own model dictionary with subtly different GPU settings.
- Prototype the SEED-IV adapter against a tiny synthetic fixture before the real download, to catch the channel-order filename and missing-label-file issues without needing the full dataset in hand first.
- Investigate the DANN collapse before moving on to Section 4.15, rather than flagging a calmer schedule as future work and proceeding: the arousal/dominance instability recurs in SSL pretraining and AdaBN too, and understanding its root cause earlier might have informed the design of those later experiments.
- Run a smaller pilot LOSO for at least one Track B architecture (e.g. TSception, the cheapest at 8765 params) to get a sense of whether full-LOSO Track B numbers would look like Track A's LOSO calibration result before ruling it out of scope on cost alone.
- Keep a changelog for the one-off notebook-authoring scripts mentioned in Appendix 5, so the provenance of any given notebook cell is easier to trace.

Chapter 12

Challenges Faced

The items below differ from Chapter 11's hindsight list in one important way: each of these was a genuine obstacle encountered *during* the work, not a process choice visible only in retrospect, and each one materially changed either the implementation or how a result had to be interpreted once it surfaced. I record them in enough technical detail to be independently verifiable, rather than as a general narrative of difficulty, because several of them (the DANN instability and the inconsistent GPU utilisation in particular) directly explain numbers that appear elsewhere in this report and would otherwise look unexplained.

1. DEAP's binarised labels are class-imbalanced in a way that misleads accuracy-only reporting, and this was not obvious until the class-balance chart and Discussion section were cross-checked against the real numbers: valence (79/21) is more skewed than my own working assumption (liking) had suggested, which changed how several later results needed to be read (e.g. Track B's 83.5% valence accuracy at 0.56 AUC).

2. scikit-learn has no built-in repeated-group cross-validation, so I had to implement my own repeated-fold generator via subject-ID relabeling, since `GroupKFold`'s greedy assignment is sensitive to the sorted order of group labels. This had to be verified by hand (zero subject overlap between train/test in every fold).

3. DANN training instability. The subject-adversarial experiment initially failed for arousal/dominance; I found and fixed an implementation bug (a validation-loader tuple-unpacking mismatch), and the collapse pattern persisted afterward, which suggests it is a real training-dynamics phenomenon matching a documented failure pattern in the literature, rather than a bug.

4. Overlapping-window non-independence. 50%-overlap windows share up to half their raw samples with their neighbours; subject grouping alone does not make within-subject windows independent, so any variance or confidence interval computed on the overlap dataset had to be caveated as optimistic rather than treated as free extra data.

5. SEED-IV access and format friction: the channel-order file is named differently and capitalised differently than the adapter first assumed, and the dataset ships no separate per-session label file, only prose in a README, requiring a defensive fallback checked by hand against that prose.

6. Inconsistent GPU utilisation across the codebase, found by auditing every script's XGBoost instantiation: the main notebook's own Track A cell requests the GPU device explicitly, but most of the Track A companion scripts default to CPU because they set a tree method without also setting the CUDA device, an easy mistake to make with the XGBoost API and easy to miss without a systematic audit.

7. The literature-attribution file referenced across the codebase and notebook prose had to be reconstructed by reading all 32 papers directly (Section 6.2); it now lives at the project root rather than inside `papers_EEG/`, which the original docstrings still point to, a minor inconsistency worth resolving.

Chapter 13

Conclusion

The central finding of this project is that subject-independent binary emotion classification on DEAP, under a leakage-free `GroupKFold`/`GroupShuffleSplit` protocol, is close to chance level (ROC-AUC 0.288–0.607) across ten different models spanning classical ML and deep learning, and that this ceiling survives a wide set of interventions: eight or more independent hypotheses tested and ruled out. A trial-within-subject permutation test confirms this directly: under a null distribution that respects each trial's 15 correlated windows and each subject's true class balance, valence, arousal, and liking's observed AUC is statistically indistinguishable from chance, and dominance's small apparent excess ($p = 0.0199$) does not survive a Bonferroni correction for testing four tasks (Section 4.14.1). An earlier, buggy version of this test that shuffled window-level labels independently understated the null's spread and overstated every effect, including a since-retracted claim that three tasks sat significantly *below* chance; that claim does not survive the corrected test and is noted here for transparency rather than silently dropped. The strongest positive result I found is Euclidean Alignment (mean AUC +0.044, Section 4.15.1), and the most important single addition is the cross-corpus validation against SEED-IV (Section 4.10), where the identical pipeline reaches a moderate AUC of 0.673 on a different corpus under a comparable protocol. That is one 3-setting comparison with a single model, so I read it as evidence that the near-chance ceiling may be specific to DEAP rather than a general property of subject-independent EEG emotion decoding, not as a result that rules out the general case. That contrast helps explain why my own numbers sit so far below the 0.97 ROC-AUC reported on the same dataset by Hasan & Islam (2025). The literature corpus includes 19 reviewed-but-unused papers alongside the 13 that drove an implemented experiment (Chapter 6); both groups are kept and labelled accordingly.

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